Algebraic Groups

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1 Preliminaries

1.1 Varieties

In this section, we denote $k = \overline{k}$, an algebraically closed field. We will mostly be working with $k = \mathbb{C}$.

Definition 1.1. *Let $\mathbb{A}^n = k^n$ be the affine n-space. We denote $\mathcal{O}(\mathbb{A}^n) = k[T_1, \cdots, T_n]$, the ring of polynomial functions on $\mathbb{A}^n$.*

Definition 1.2. *Let $S \subseteq \mathcal{O}(\mathbb{A}^n)$ be a subset. We define the vanishing set of S to be $V(S) = \{x \in \mathbb{A}^n \mid \forall f \in S, f(x) = 0\}$.*

Remark 1.1. *We have, $V(S) = V((S))$ where (S) is an ideal generated by S. Moreover, for any ideal I we have,*

$$V(I) = V\left(\sqrt{I}\right).$$

Definition 1.3. *We introduce a topology on $\mathbb{A}^n$ by setting $\{V(I)\}_{I \subseteq \mathcal{O}(\mathbb{A}^n) \text{ ideals.}}$ to be the closed sets. We call this topology the Zariski topology.*

Remark 1.2. *All points in $\mathbb{A}^n$ are closed. Indeed, for $a \in \mathbb{A}^n$, and the ideal $\mathfrak{m}_a = (x_1 - a_1, \cdots, x_n - a_n)$. We have $V(\mathfrak{m}_a) = \{a\}$.*

Definition 1.4. *Let $X \subseteq \mathbb{A}^n$, then we define*

$$I(X) = \{f \in \mathcal{O}(\mathbb{A}^n) \mid \forall x \in X, f(x) = 0\}.$$

Theorem 1.1 (Hilbert Nullstellen Satz)**.** *For all $X \subset \mathbb{A}^n$ and $I \subseteq \mathcal{O}(\mathbb{A}^n)$ ideal, we have*

$$V(I(X)) = \overline{X}, \quad I(V(I)) = \sqrt{I}.$$

Corollary 1.1. *Both function I, V reverse inclusion relations. Furthermore, they are inverse to one another between the sets*

$$\{S \subseteq \mathbb{A}^n \mid S = \overline{S}\} \underset{V}{\overset{I}{\leftrightarrows}} \{I \subseteq \mathcal{O}(\mathbb{A}^n) \mid I = \sqrt{(I)}\}$$

Corollary 1.2. *We have the following bijections.*

$$\mathbb{A}^n \overset{I}{\leftrightarrow} \mathrm{MaxSpec}(\mathcal{O}(\mathbb{A}^n)) \leftrightarrow \mathrm{Hom}_{\mathrm{k-alg}}(\mathcal{O}(\mathbb{A}^n), k),$$

where $\mathrm{MaxSpec}(\mathcal{O}(\mathbb{A}^n))$ is the set of maximal ideals and $\mathrm{Hom}_{\mathrm{k-alg}}(\mathcal{O}(\mathbb{A}^n), k)$ is the set of all evaluation maps.

Proof. For the first bijection, $\mathbb{A}^n \to \mathrm{MaxSpec}(\mathcal{O}(O)\mathbb{A}^n)$ is given by Remark1.2. It remains to show that I maps points to maximal ideals. Since I reverses the inclusion and $\{a\}$ is minimal closed set, $I(\{a\})$ is the maximal ideal which contains $\mathfrak{m}_a$. By the maximality of $\mathfrak{m}_a$, they are equal.

For the second bijection, we observe that for any $\varphi \in \mathrm{Hom}_{\mathrm{k-alg}}(\mathcal{O}(\mathbb{A}^n), k)$,

$$\mathcal{O}(\mathbb{A}^n)/\ker\varphi \cong k.$$

This means that $\ker\varphi$ is maximal ideal. Since k is algebraically closed, all the maximal ideal is generated by n many degree 1 polynomials. We conclude

$$\varphi = \mathrm{ev}_a$$

for some $a \in \mathbb{A}^n$. □

Definition 1.5. *A concrete affine variety is a closed subset X of some affine space $\mathbb{A}^n$ equipped with the subspace topology of the Zariski topology on $\mathbb{A}^n$.*

Definition 1.6. *Let $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$ be affine varieties. A morphism of affine varieties between X and Y is a map $f : X \to Y$ such that there exist $f_1, \cdots, f_m \in \mathcal{O}(\mathbb{A}^n)$ with*

$$f(x) = (f_1(x), \cdots, f_m(x)).$$

A morphism of affine varieties is called an isomorphism if there exist another morphism of affine varieties $g : Y \to X$ satisfying

$$g \circ f = \mathrm{id}_X, \quad f \circ g = \mathrm{id}_Y .$$

Example 1.1. *Following subsets of $\mathbb{A}^n$ are affine varieties.*

1). $\mathbb{A}^n$ itself as $\mathbb{A}^n = V(\emptyset)$.

2). A finite set of points $\{a_1, \cdots, a_m\} = V(\mathfrak{m}_{a_1} \cdots \mathfrak{m}_{a_m})$.

3). Any finite union and any possibly infinite intersection of affine varieties are again affine varieties.

Definition 1.7.
Let $X = V(S) \subseteq \mathbb{A}^n, Y = V(T) \subseteq \mathbb{A}^m$. We define $X \times Y = V(S \coprod T) \subseteq \mathbb{A}^{n+m}$ where we regard,

$$S \subseteq k[x_1, \cdots, x_n] \subseteq k[x_1, \cdots, x_{n+m}]$$

and

$$T \subseteq k[x_{n+1}, \cdots, x_{n+m}] \subseteq k[x_1, \cdots, x_{n+m}].$$

Observe that $X \times Y$ is an affine variety.

Lemma 1.1. *A morphism $f : X \to Y$ of affine varieties is continuous with respect to Zariski topology.*

Proof. Since X, Y have the subspace topologies, we can assume $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$. And the continuity amounts to show that any closed subsets of $\mathbb{A}^m$ has a closed preimage. Let $S \subseteq \mathcal{O}(\mathbb{A}^m)$, then by the definition of morphisms of affine varieties,

$$a \in f^{-1}(V(S)) \Rightarrow \forall \varphi \in S, \varphi \circ f(a) = 0.$$

Therefore

$$f^{-1}(V(S)) = V(\{\varphi \circ f \mid \varphi \in S\}).$$

□

Definition 1.8. *Let $X \subseteq \mathbb{A}^n$ be an affine variety. The ring of regular polynomial functions on X is*

$$\mathcal{O}(X) := \mathcal{O}(\mathbb{A}^n)/I(X).$$

Proposition 1.1. *The ring of regular polynomial functions is a finitely generated k-algebra. If we consider the set $\mathrm{Map}(X, k)$ of maps from $X \to k$ as a k-algebra by letting point-wise sum and product as the addition and the multiplication, then $\mathcal{O}(X)$ can be viewed as a sub-algebra of $\mathrm{Map}(X, k)$.*

Proof. The first claim is obvious as $\mathcal{O}(\mathbb{A}^n)$ is finitely generated. For the second claim, let us consider the evaluation map

$$\mathcal{O}(\mathbb{A}^n) \ni f \mapsto [x \mapsto f(x)] \in \mathrm{Map}(X, k).$$

Then its kernel is $I(X)$. Therefore,

$$\mathcal{O}(X) \hookrightarrow \mathrm{Map}(X, k).$$

□

Definition 1.9. *A topological space X is called irreducible if it cannot be written as a union of two proper closed subsets.*

Example 1.2. *The only irreducible sets in $\mathbb{R}^n$ are the points.*

Lemma 1.2. *We have the following statements*

(1). An affine variety X is irreducible if and only if $\mathcal{O}(X)$ is an integral domain.

(2). An affine variety X is connected if and only if $\mathcal{O}(X)$ does not contain a non-trivial idempotent.

Proof. Suppose $\mathcal{O}(X)$ is not an integral domain, then there are non-zero zero-divisors $f, g \in \mathcal{O}(X)$ such that $fg = 0$. Observe that,

$$V(f) \cup V(g) = V(fg) = V(0) = X.$$

As f, g are not 0, X is not irreducible. The converse also follows as all closed subsets are of the form $V(I)$ and V reverses the inclusions.

For the second assertion, let $e \in \mathcal{O}(X)$ be an idempotent then $e(1-e) = 0$. Therefore $X = D(e) \sqcup D(1-e)$. If e can be taken as a non-trivial element, then X is disconected. For the converse, suppose $\mathcal{O}(X)$ does not admit a non-trivial idempotent but X is disconnected thus $X = U \sqcup V$ for some proper open sets. Then there are $I, J \subseteq k[x_1, \cdots, x_n]$ such that $X \cap V(I) = U, X \cap V(J) = V$. Since U, V are disjoint, $I + V = k[x_1, \cdots, x_n]$. Therefore there is $a \in I, b \in J$ with $a + b = 1$. Note that $ab = 0$ therefore $b^2 = b(1-a) = b$. In particular $a, b \in \{0, 1\}$. □

Definition 1.10. *Let X be an affine variety and $a \in X$. We define the maximal ideal $\mathfrak{m}_a$ to be such that*

$$\mathfrak{m}_a := \{f \in \mathcal{O}(X) \mid f(x) = 0\} = \ker(\mathrm{ev}_a : \mathcal{O}(X) \to k).$$

Corollary 1.3. *For an affine variety X, we have the following bijections.*

$$X \leftrightarrow \mathrm{MaxSpec}(\mathcal{O}(X)) \leftrightarrow \mathrm{Hom}_{\mathrm{k-alg}}(\mathcal{O}(X), k).$$

Proof. Observe that we have

$$X = V(I(X))$$

and

$$\mathrm{MaxSpec}(\mathcal{O}(X)) = \{\mathfrak{m} \in \mathrm{MaxSpec}(\mathcal{O}(\mathbb{A}^n)) \mid I(X) \subseteq \mathfrak{m}\}.$$

Since V is inclusion reversing.

$$V(\mathfrak{m}_a) = \{a\} \subseteq V(I(X)) \Leftrightarrow \mathfrak{m}_a \subseteq I(X).$$

For the second bijection, obviously any kernel of k-algebra homomorphism $\varphi : \mathcal{O}(X) \to k$ is maximal. □

Definition 1.11. *An (abstract) affine variety is a pair* $(X, \mathcal{O}(X))$ *where* X *is a set and* $\mathcal{O}(X)$ *is a finitely generated* k*-subalgebra of* $\mathrm{Map}(X, k)$ *such that*

$$X \to \mathrm{Hom}_{\mathrm{k-alg}}(\mathcal{O}(X), k), \quad x \mapsto \mathrm{ev}_x$$

is a bijection.

Definition 1.12. *A morphism of abstract affine varieties is a map* $f : X \to Y$ *such that the map*

$$f^{\#} : \mathrm{Map}(Y, k) \to \mathrm{Map}(X, k), \quad \varphi \mapsto \varphi \circ f$$

restricts to an k*-algebra homomorphism*

$$f^{\#} : \mathcal{O}(Y) \to \mathcal{O}(X).$$

Definition 1.13. *A ring is reduced if* 0 *is the only nilpotent of it.*

Remark 1.3. *Note that* k *is a field therefore reduced, we have* $\mathrm{Map}(X, k)$ *is reduced as a* k*-algebra and so is* $\mathcal{O}(X)$*.*

Notation 1.1. *We denote the categories of concrete and abstract affine varieties as* ConcAffVar *and* AbsAffVar.

Lemma 1.3. *Any concrete affine variety* X *defines an abstract affine variety* $(X, \mathcal{O}(X))$*. Furthermore, this gives an equivalence of categories*

$$(\mathrm{ConcAffVar})X \xrightarrow{\sim} (X, \mathcal{O}(X))(\mathrm{AbsAffVar}).$$

Proof. First we need to show that the functor is well-defined. Indeed with Definition 1.8, we already know that $\mathcal{O}(X)$ is a finitely generated k-subalgebra of $\mathrm{Map}(X, k)$. Furthermore, a morphism of concrete affine varieties $f : X \to Y$ gives a morphism between $\mathcal{O}(Y) \to \mathcal{O}(X)$ by

$$\mathcal{O}(Y) \ni g \mapsto g \circ f \in \mathcal{O}(X).$$

This is faithful by looking at the images of $y_1, \cdots, y_n$ where $Y \subseteq \mathbb{A}^n_k$ under $f^{\#}$. Given $\varphi : \mathcal{O}(Y) \to \mathcal{O}(X)$, set

$$f(x) = (\varphi(y_1)(x), \cdots, \varphi(y_n)(x)),$$

then this gives a morphism of concrete affine varieties.

Finally for the essentially surjectivity, it is clear that for $\mathcal{O}(X) \cong k[x_1, \cdots, x_n]/I$, then $V(I) \subseteq \mathbb{A}^n_k$ gives the right preimage for the finitely generated k-algebra part. On the set part, it is clear from Corollary 1.3. □

From now on, we will refer to both concrete and abstract varieties simply as affine varieties.

Remark 1.4. *Let X be an affine variety. Then $\mathcal{O}(X)$ is Noetherian since it is a finitely generated k-algebra.*

Definition 1.14. *A topological space X is Noetherian if any descending chain of closed subsets eventually stabilizes.*

Remark 1.5. *An affine variety is Noetherian.*

Definition 1.15. *A maximal closed irreducible subset of a topological space is called its irreducible component.*

Lemma 1.4. *Any Noetherian topological space X can be written as a finite union*

$$X = X_1 \cup \cdots \cup X_m$$

where each X_i is closed and irreducible with $X_i \subseteq X_j \Rightarrow i = j$ (ie. they are irreducible components of X). Furthermore, such $(X_i)_{i=1,\cdots,m}$ is unique up to reordering.

Proof. Follows directly from the definition of Noetherian spaces. □

Example 1.3. *Consider $X = V(x_1, x_2) \subseteq \mathbb{A}^2$. Then $X = \mathbb{A}^1 \times \{0\} \cup \{0\} \times \mathbb{A}^1$ where each factor is an irreducible component.*

Lemma 1.5. *Let X be irreducible topological space and $U \subseteq X$ be a non-empty open subset. Then U is dense in X.*

Proof. We can decompose $X = \overline{U} \cup (X \backslash U)$ where each factor is closed. Since $X \backslash U \neq X$, we conclude $\overline{U} = X$. □

1.2 Algebraic Groups

Definition 1.16. *Let X be a topological space. The dimension* $\dim X$ *of X to be*

$$\dim X = \max \left\{ i \in \mathbb{N} \;\middle|\; \begin{array}{c} \exists \emptyset \subsetneq V_0 \subsetneq \cdots \subsetneq V_d \subseteq X, \\ \text{s.t. } \{V_i\}_i \text{ are irreducible closed subsets} \end{array} \right\}.$$

Definition 1.17. *Let X be an affine variety. We define*

$$k(X) = \mathrm{Frac}(\mathcal{O}(X)).$$

Theorem 1.2. *Let X be an irreducible variety. Then we have*

$$\dim X = \mathrm{trdeg}(k(X)/k).$$

Example 1.4. *Let $X = \mathbb{A}^n$, we have*

$$k(\mathbb{A}^n) = \mathrm{Frac}(k[x_1, \cdots, x_n]) = k(x_1, \cdots, x_n).$$

Thus we obtain,

$$\dim \mathbb{A}^n = \mathrm{trdeg}(k(x_1, \cdots, x_n)/k) = n.$$

Example 1.5. *Let $f \in k[x_1, \cdots, x_n] = \mathcal{O}(\mathbb{A}^n)$ be non-zero element. Let $X = V(fy - 1) \subset \mathbb{A}^{n+1}$. Observe that*

$$fy - 1 \in \mathcal{O}(\mathbb{A}^n)[y] \cong \mathcal{O}(\mathbb{A}^{n+1}).$$

Therefore we have,

$$\mathcal{O}(X) = \mathcal{O}(\mathbb{A}^{n+1})/(fy - 1) \cong \mathcal{O}(\mathbb{A}^n)[f^{-1}] \subseteq \mathrm{Frac}(\mathcal{O}(\mathbb{A}^n)).$$

In particular, $k(X) = k(\mathbb{A}^n)$, therefore

$$\dim X = n.$$

Definition 1.18. *An affine algebraic group G is an affine variety together with a group structure.*

$$m : G \times G \to G, m(g, h) = gh, \quad i : G \to G, i(g) = g^{-1},$$

which are morphisms of varieties.

Definition 1.19. *A morphism of algebraic groups is a map $f : G \to H$ which is a morphism of varieties that is also a group homomorphism.*

Definition 1.20. *A closed subset $H \subseteq G$ of an algebraic group G is called a closed subgroup if it is a subgroup. In particular, a closed subgroup is again an algebraic group.*

Example 1.6. *Any finite group is an algebraic group by embedding $G \subset \mathbb{A}^1$ since finite sets are always closed.*

Remark 1.6. *Products of algebraic groups are again algebraic groups.*

Definition 1.21. *An additive group is $\mathbb{G}_a = \mathbb{A}^1$ which is a group under addition*

Definition 1.22. *A multiplicative group $\mathbb{G}_m = V(xy - 1) \subset \mathbb{A}^2$ is an algebraic group such that*

1. *the multiplication is $m : \mathbb{G}_m \times \mathbb{G}_m \to \mathbb{G}_m, (x_1, y_1)(x_2, y_2) = (x_1x_2, y_1y_2)$*
2. *the inversion is $i : \mathbb{G}_m \to \mathbb{G}_m, i(x_1, y_1) = (y_1, x_1)$.*

Remark 1.7.

$$\mathcal{O}(\mathbb{G}_m) = k[x, y]/(xy - 1) \cong k[x, x^{-1}].$$

Recall from Linear Algebra, that we have for any matrix $A \in \mathrm{Mat}_{n\times n}(k)$,

$$A \,\mathrm{Adj}(A) = \det A \cdot I_n.$$

Both Adj, det are defined in terms of polynomials. By viewing n-square matrices as a n^2-dimensional vectorspace, we define following algebraic groups which are compatible with the classical ones.

Definition 1.23. *The special linear group is*

$$\mathrm{SL}_n = \{A \in \mathrm{Mat}_{n\times n} \mid \det(A) = 1\} = V(\det(A) - 1) \subset \mathbb{A}^{n\times n}.$$

The matrix multiplication is given in polynomial form. Moreover, from above fact from Linear Algebra, we see

$$A^{-1} = \mathrm{Adj}(A).$$

Thus the inverse is also givein in polynomials.

Definition 1.24. *The general linear group is*

$$\mathrm{GL}_n := \{A \in \mathrm{Mat}_{n\times n} \mid \det A \neq 0\} = \{(A, y) \in \mathrm{Mat}_{n\times n} \times \mathbb{A}^1 \mid \det A \cdot y = 1\}.$$

The multiplication is $(A, a)(B, b) = (AB, ab)$ *thus in polynomials. The inverse is*

$$(A, a)^{-1} = (a\,\mathrm{Adj}(A), \det A).$$

Thus given in polynomials.
As an abstract variety, we have

$$(V(\det(x_{ij})_{ij} \cdot y - 1), k[x_{ij}]_{ij}[\det(x_{ij})^{-1}]).$$

Definition 1.25. *Let* $J \in \mathrm{GL}_n$ *then we define an algebraic group,*

$$G(J) = \{A \in \mathrm{Mat}_{n\times n} \mid A^t J A = J\}.$$

Proposition 1.2. *Let* $J \in \mathrm{Mat}_{n\times n}$ *be a symmetric matrix. Then we have,*

$$G(J) \cong \mathrm{O}_n\,.$$

Proof. Any symmetric matrix of complex coefficients are congruent to one another. □

Proposition 1.3. *Let* $J \in \mathrm{Mat}_{n\times n}$ *be a skew symmetric matrix. Then we have,*

$$G(J) \cong \mathrm{Sp}_{2n}\,.$$

Remark 1.8. *It can be shown that for any* $J \in \mathrm{GL}_n$ *we have,*

$$G(J) \cong \mathrm{O}_n \text{ or } \mathrm{Sp}_{2n}\,.$$

Definition 1.26. *If* $H \subseteq G$ *is a closed subgroup, then we define the centralizer of* G *to be*

$$C_G(H) := \{x \in G \mid \forall h \in H\, xh = hx\}.$$

and the normalizer of G *to be*

$$N_G(H) := \{x \in \mathbb{G} \mid xHx^{-1} = H\}.$$

Proposition 1.4. $C_G(H), N_G(H)$ *are closed subgroups of* G.

Proof. We will only give a proof for the centralizers. Let $h \in G$ then

$$C_G(\{h\}) = c_h^{-1}(\{h\})$$

where

$$c_h(g) = ghg^{-1}.$$

c_h is indeed a morphism of algebraic groups as,

$$c_h(\cdot) = m(m(\cdot, h), i(\cdot)).$$

Furthermore,

$$C_G(H) = \bigcap_{h\in H} C_G(\{h\}),$$

thus closed. □

Remark 1.9. *Any connected* 1*-dimensional affine algebraic group is isomorphic to either* $\mathbb{G}_a$ *or* $\mathbb{G}_m$*.*

Theorem 1.3. *Any affine algebraic groups is a closed subgroup of* GL_n *for some* $n \in \mathbb{N}$*.*

Proof. Foundation course. □

Example 1.7. *We give explicit embedding of the above theorem for following two groups.*

1. $\mathrm{SL}_n \hookrightarrow \mathrm{GL}_n, A \mapsto (A, 1)$.
2. $\mathbb{G}_a \hookrightarrow \mathrm{GL}_2, a \mapsto \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$.

Definition 1.27. *Let* G *be an algebraic group.* G^0 *denotes the connected component of* G *which contains the identity.*

Lemma 1.6. *For an algebraic group* G*, we have*

1). G^0 *is a closed normal subgroup of finite index.*

2). For any $g \in G$*, the coset* gG^0 *is connected and irreducible component of* G*.*

3). Each closed subgroup H *of* G *of finite index contains* G^0*. In other words,* G^0 *is a minimal subgroup of finite index.*

Lemma 1.7. *Let* $\varphi : G \to H$ *be a morphism of algebraic groups. We have,*

1). $\ker \varphi, \operatorname{im} \varphi$ *are closed subgroups.*

2). $\varphi(G^0) = \varphi(G)^0$.

3). $\dim G = \dim \ker \varphi + \dim \operatorname{im} \varphi$.

Proof. Foundation course. □

Definition 1.28. *Let* G *be an algebraic group and* X *be a variety. An algebraic action of* G *on* X *is an abstract group action such that the action map*

$$a : G \times X \to X$$

is a morphism of varieties.

Definition 1.29. *Let* G *be an algebraic group. A finite dimensional representation of* G *is an algebraic action of* G *on* $\mathbb{A}^n$ *for some* $n \in \mathbb{N}$ *which is* k*-linear.*

Definition 1.30. *A rational represenattion of an algebraic group* G *is a representation* (π, V) *of* G *(as an abstract group) such that* V *is a union of finite dimensional algebraic subrepresentations.*

From now on, we refer an algebraic representation simply as a representation.

Example 1.8. *Linear transformations* $\mathrm{GL}_n \times k^n \to k^n$ *is a finite dimensional representations. Similarly for* $\mathrm{SL}_n, \mathrm{O}_n, \mathrm{Sp}_{2n}$ *and the symmetric group* $\mathfrak{S}_n$ *as transpositions can be expressed using matrices.*

Example 1.9. *If* V *is a* G*-representation then* V^* *is a* G*-representation via,*

$$\forall f \in V^*, g \in G, v \in V, (g \cdot f)(v) = f(g^{-1}v).$$

Example 1.10. *Let V, W be G-representations, then $V \otimes_k W$ is a G-representation via.*

$$\forall g \in G, v \in V, w \in W, g(v \otimes w) = gv \otimes gw.$$

Definition 1.31. *A morphism of G-representations is a k-linear map $f : V \to W$ such that*

$$\forall g \in G, v \in V, g \cdot f(v) = f(g \cdot v).$$

Example 1.11. *For a morphism $f : V \to W$ of G representation,* $\ker f, \operatorname{im} f, \operatorname{coker} f$ *are sub-G-representations.*

Example 1.12. *Let G_1, G_2 be algebraic groups and V_1, V_2 be representations of G_1, G_2, respectively. We define a $G_1 \times G_2$-representation via*

$$\forall v_1 \in V_1, v_2 \in V_2, \forall (g_1, g_2) \in G_1 \times G_2, (g_1, g_2)(v_1 \otimes v_2) = g_1 v_1 \otimes g_2 v_2.$$

We call this an exterior product of V_1 and V_2 for $G_1 \times G_2$-representation and denote it as $V_1 \boxtimes V_2$.

Definition 1.32. *We can consider an action of $G \times G$ on $\mathcal{O}(G)$ via*

$$((g_1, g_2)f)(x) = f(g_1^{-1} x g_2).$$

Remark 1.10. *Above representation is well-defined (shown in the foundation course). Furthermore, this is a rational $G \times G$-representation.*

Definition 1.33. *A representaion is said to be semi-simple if it is a direct sum of irreducible representations.*

Definition 1.34. *An algebraic group G is called linearly reductive if the category of representations* $\operatorname{Rep}(G)$ *of G is semi-simple (i.e. any finite dimensional representation of G is semi-simple).*

Definition 1.35 (Vandermonde Matrix)**.** *A Vandermonde matrix is a function $V : \mathbb{R}^{m+1} \to \operatorname{Mat}_{m+1 \times n+1}$ such that*

$$V(x_0, \cdots, x_m) = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^n \end{pmatrix}.$$

Proposition 1.5. *For $m = n$ in the above case we have*

$$\deg(V(x_0, \cdots, x_m)) = \prod_{0 \le i < j \le m} (x_j - x_i).$$

Proposition 1.6. *$\mathbb{G}_m$ is linearly reductive. Furthermore, there is an irreducible $\mathbb{G}_m$-representation k_n of dimension 1 for each $n \in \mathbb{Z}$ such that*

$$\forall v \in k_n, x \in \mathbb{G}_m, x \cdot v = x^n v.$$

The set $\{k_n \mid n \in \mathbb{Z}\}$ is a complete list of pairwise non-isomorphic irreducible $\mathbb{G}_m$-representations.

Proof. Recall that every element of $\mathbb{G}_m$ is of the form (x, x^{-1}) for some $x \in k^\times$. Let V be a $\mathbb{G}_m$-representation and also we denote

$$V_n = \{v \in V \mid \forall x \in \mathbb{G}_m, (x, x^{-1}) \cdot v = x^n v\},$$

We will first show that

$$V = \bigoplus_{n \in \mathbb{Z}} V_n.$$

Indeed, let $v \in V_n \cap V_m$, then

$$\forall x \in \mathbb{G}_m, x^n v = x^m v.$$

Thus $v = 0$. By a similar argument, we get, for any finite set $S \subseteq \mathbb{Z}$ and $n \in \mathbb{Z}$,

$$\left(\sum_{m \in S} V_m\right) \cap V_n = \{0\}.$$

Now we want to show

$$\sum_{n \in \mathbb{Z}} V_n = V.$$

Let us fix $v \in V$, then we get a morphism of varieties

$$\mathbb{G}_m \ni x \mapsto xv \in V.$$

By definition, $\rho : \mathbb{G}_m \times V \to V$ is expressed in terms of polynomial so is $\rho(\cdot, v) : \mathbb{G}_m \to V$. Thus there exists $f_1(t), \cdots, f_n(t) \in \mathcal{O}(\mathbb{G}_m) = k[t, t^{-1}]$ such that

$$tv = \sum_{i=1}^{n} f_i(t) e_i,$$

where $\{e_i\}_{i=1,\cdots,n}$ is the standard basis. The above equation can also be written as

$$tv = \sum_{i=-N}^{M} t^i v_i,$$

for some $v_n \in V$.We want to show $v_i \in V_i$ for each $-N \leq i \leq M$.

For any $s, t \in \mathbb{G}_m$, we have

$$st \cdot v = \sum_{i=-N}^{M} (st)^i v_i = \sum_{i=-N}^{M} t^i (s^i v_i).$$

On the other hand,

$$st \cdot v = s \cdot (t \cdot v) = s \cdot \left(\sum_{i=-N}^{M} t^i v_i\right) = \sum_{i=-N}^{M} t^i (s \cdot v_i).$$

Let $t_1, \cdots, t_{N+M+1} \in \mathbb{G}_m$ be distinct, then

$$\begin{pmatrix} t_1^{-N} & \cdots & t_1^{M} \\ \vdots & \ddots & \vdots \\ t_{N+M+1}^{-N} & \cdots & t_{N+M+1}^{M} \end{pmatrix} \begin{pmatrix} s^{-N} v_{-N} - s \cdot v_{-N} \\ \vdots \\ s^{M} v_M - s \cdot v_M \end{pmatrix} = 0.$$

By Proposition 1.5, we get

$$\det \begin{pmatrix} t_1^{-N} & \cdots & t_1^{M} \\ \vdots & \ddots & \vdots \\ t_{N+M+1}^{-N} & \cdots & t_{N+M+1}^{M} \end{pmatrix} = (t_1 \cdots t_{N+M+1})^{-N} \prod_{i<j} (t_j - t_i) \neq 0.$$

Thus we conclude for each s

$$s \cdot v_i = s^i v_i.$$

In particular, for any $v \in V$,

$$v = 1 \cdot v = \sum_{i=-N}^{M} v_i.$$

Let $w \in V_i$ then we have for any $t \in \mathbb{G}_m$,

$$t \cdot w = t^n w.$$

Therefore, $V_i = (k_i)^{\dim V_i}$. Any representation of $\mathbb{G}_m$ is a direct sum of k_i-s. □

Proposition 1.7. *Let V be a finite dimensional vector space and G be an algebraic group. Then any morphisms algebraic grouops*

$$\rho : G \to \mathrm{GL}(V)$$

defines a representation via

$$G \times V \xrightarrow{\rho} \mathrm{GL}(V) \times V \to V.$$

Furthermore, this defines a bijection,

$$\mathrm{Mor}(G, \mathrm{GL}(V)) \leftrightarrow \{G\text{-representations on } V\}.$$

From now on, we will freely use both notions interchangebly.

Proposition 1.8. *The group $\mathbb{G}_a$ has a unique irreducible representation, namely the trivial representation. Furthermore, $\mathbb{G}_a$ is not linearly reductive.*

Proof. Let V be a $\mathbb{G}_a$-representation corresponding to $\rho : G \to \mathrm{GL}(V)$. Explicitly,

$$\mathbb{G}_a \xrightarrow{\rho} \mathrm{GL}(V) \subseteq \mathrm{End}(V) \cong \mathbb{A}^{(\dim V)^2}.$$

Pick a basis $\{v_1, \cdots, v_n\} \subset V$.

By the definition of morphisms of varieties, we have

$$\rho(t) = \sum_{1 \leq i,j \leq n} f_{ij}(t) E_{ij}$$

for some polynomials $\{f_{ij}\}_{1\leq i,j\leq n}$. Order by the degree of the monomials we get

$$\rho(t) = \sum_{i=0}^{\infty} t^i \varphi_i$$

for some $\{\varphi_n\} \subset \mathrm{End}(V)$, which is independent of the choice of element t. Note that for $n >> 0$, $\varphi_n = 0$. By plugging in $t = 0 \in G_a$, we get

$$v = 0 \cdot v = \varphi_0(v),$$

hence $\varphi_0 = \mathrm{id}_V$. Let $s, t \in \mathbb{G}_a$, and consider

$$s \cdot (t \cdot v) = s \cdot \left(\sum_{n=0}^{\infty} t^n \varphi_n(v) \right) = \sum_{n=0}^{\infty} t^n s \varphi_n(v) = \sum_{n,m=0}^{\infty} t^n s^m \varphi_m \circ \varphi_n(v).$$

Since $G_a \curvearrowright V$, thus we have

$$s \cdot (t \cdot v) = (s + t) \cdot v.$$

From this we get

$$(s + t) \cdot v = \sum_{n,m=0}^{\infty} \binom{n+m}{n} s^m t^n \varphi_{m+n}(v).$$

Using Vandermonde's matrix again, we see,

$$\binom{n+m}{n} s^m t^n \varphi_{m+n}(v) = t^n s^m \varphi_m \circ \varphi_n(v).$$

Thus for any $v \in V$, we have

$$\binom{n+m}{n} \varphi_{m+n}(v) = \varphi_m \circ \varphi_n(v).$$

We will use this fact to compute $(t-1)^k v$. We will prove the following assertion.

$\forall l \geq 1$, we have

$$(t-1)^l v = \sum_{n=l}^{\infty} p_n^{(l)}(t) \varphi_n(v),$$

where $p_n^{(l)}(t) \in k[t]$.

Indeed for $l = 1$, we had $\varphi_0 = \mathrm{id}_V$. For the induction part $l \to l + 1$, we have

$$\begin{aligned}
(t-1)^{l+1} v &= (t-1)((t-1)^l v), \\
&= (t-1) \left(\sum_{n=l}^{\infty} p_n^{(l)}(t) \varphi_n(v) \right), \\
&= \left(\sum_{n=l}^{\infty} p_n^{(l)}(t) t^n \varphi_n(v) \right) - \left(\left(\sum_{n=l}^{\infty} p_n^{(l)}(t) \varphi_n(v) \right) \right), \\
&= \sum_{n=l+1}^{\infty} p_n^{(l+1)}(t) \varphi_l(v),
\end{aligned}$$

where $p_n^{(l+1)}(t) = p_n^{(l)}(t)(t^n - 1)$.

Recall that $\varphi_n = 0$ for $n >> 0$. The above discussion shows us that

$$n >> 0 \Rightarrow (t-1)^n v = 0.$$

Let $v \neq 0$ and m be the maximum of such that $(t-1)^m v \neq 0$. Let $w = (t-1)^m v$. Then

$$(t-1)w = 0 \Rightarrow \forall t \in G_a, tw = w.$$

In other words, $\langle w \rangle \subset V$ is a copy of the trivial representation. Therefore any represenattion contains the trivial representation as a subrepresentation. In particular, V is irreducible if and only if it is the trivial representation.

To see that $\mathbb{G}_a$ is not linearly reductive. Consider,

$$\mathbb{G}_a \ni t \mapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2 .$$

The only $\mathbb{G}_a$ stable subspaces are

$$\{0\}, \langle e_1 \rangle, k^2.$$

But $\langle e_1 \rangle$ does not admit a $\mathbb{G}_a$-stable complement in k^2. Hence k^2 is not semisimple. □

Theorem 1.4 (Schur)**.** *Let G be a group and V, W be finite irreducible G-representations over a field k.*

1. $\mathrm{Hom}_G(V, W)$ *is* 1*-dimensional if* $V \cong W$*, otherwise it is* $\{0\}$.
2. *If k is algebraically closed then* $\mathrm{End}_G(V) \cong k \,\mathrm{id}_V$.

Theorem 1.5 (Artin-Wedderburn)**.** *For a group G, let $V = V_1 \oplus \cdots V_r$ be a semisimple G-reprensetation. Then we have,*

$$\mathrm{End}_G(V) \cong \bigoplus_{i=1}^{r} \mathrm{End}_G(V_i).$$

Theorem 1.6 (Burnside)**.** *Let k be an algebraically closed field, V be a finite dimensional k-vectorspace and $A \subseteq \mathrm{End}_k(V)$ be a unital k-subalgebra. If V is an irreducible A-representation by the canonical matrix multiplication action to vectors, then $A = \mathrm{End}_k(V)$.*

Lemma 1.8. *Let G_1, G_2 be affine algebraic groups then there exists a bijection between*

$$\mathrm{Irr}(G_1) \times \mathrm{Irr}(G_2) \leftrightarrow \mathrm{Irr}(G_1 \times G_2),$$

such that

$$(V_1, V_2) \mapsto (V_1 \boxtimes V_2).$$

Proof. Let A_1, A_2 be the images of G_1, G_2 under the representation. By Theorem 1.6, they conincide with $\mathrm{End}_k(V_1), \mathrm{End}_k(V_2)$. Let A be the image of $G_1 \times G_2$ under the representation then $A_1 \otimes A_2 \subseteq A$. By counting the dimension, we conclude they are isomorphic. Thus $\rightarrow$ is justified.

For the other direction, let V be an irreducible $G_1 \times G_2$-representation. Let $V_1 \subseteq V$ be an irreducible $G_1 \cong G_1 \times \{e_2\}$-subrepresentation of V. Since action by G_1 commutes

with action by G_2, hV_1 is also an irreducible subrepresentation of V where $h \in G_2$. Using the irreducibility of V,

$$V = \sum_{h \in G_2} hV_1.$$

Let $V_2 = \mathrm{Hom}_{G_1}(V_1, V)$ and define G_2-representation on V_2 by

$$G_2 \times V_2 \ni (h, f) \mapsto [v \mapsto hf(v)] \in M.$$

Again by the fact that G_1 and G_2 commute in $G_1 \times G_2$, this is well-defined. Furthermore, V_2 is finite dimensional as it is contained in a finite dimensional space $\mathrm{Hom}_k(V_1, V)$. The evaluation map,

$$V_1 \boxtimes V_2 \ni v \otimes f \overset{\Phi}{\mapsto} f(v) \in V,$$

is then $G_1 \times G_2$-equivariant. This is clearly non-zero hence by Theorem 1.4, Φ is surjective. Furthermore, each $f \in V_2$ is injective unless it is 0, hence Φ is injective. We conclude $V_1 \boxtimes V_2 \cong V$. For any G_2-subrepresentation $W \subseteq V_2$ gives a subrepresentation $V_1 \boxtimes W \subseteq V$. By the irreducibility of V, we see W is trivial. □

Lemma 1.9. *If G_1, G_2 are linearly reductive so is $G_1 \times G_2$.*

Proof. Motivated by the proof of Lemma 1.8, let $V \in \mathrm{Rep}(G_1 \times G_2)$ and consider,

$$W = \bigoplus_{U \subseteq \mathrm{Irr}(G_1)} U \boxtimes \mathrm{Hom}_{G_1}(U, V).$$

In the proof of Lemma 1.8, we have seen that $U \boxtimes \mathrm{Hom}_{G_1}(U, V)$ is injective. Hence the evaluation map from W to V is injective as it is a direct sum of injective maps. Therefore $W \cong V$. Using the linearly reductivity of G_2 on $\mathrm{Hom}_{G_1}(U, V)$, we have shown the statement. □

We have seen that all irreducible $\mathbb{G}_m$ representations are 1-dimensional. Hence we have a bijection,

$$\mathrm{Irr}(\mathbb{G}_m) \leftrightarrow \mathrm{Mor}_{\text{Alg-Grps}}(\mathbb{G}_m, \mathbb{G}_m).$$

Definition 1.36. *Let G be an algebraic group. A character of G is a morphism of algebraic group*

$$G \to \mathbb{G}_m = \mathrm{GL}_1 .$$

The set of all characters is denoted by $X^*(G)$.

Remark 1.11. *$X^*(G)$ forms a group via*

$$\lambda, \mu \in X^*(G), (\lambda \cdot \mu)(g) = \lambda(g)\mu(g).$$

Lemma 1.10. *Let G_1, G_2 be affine algebraic gruops. Then*

$$X^*(G_1 \times G_2) = X^*(G_1) \times X^*(G_2).$$

Proof. Let us consider the following maps,

$$C_1 : G_1 \hookrightarrow (G_1 \times G_2), g \mapsto (g, e_{G_2}),$$

and

$$C_2 : G_2 \hookrightarrow (G_1 \times G_2), g \mapsto (e_{G_1}, g).$$

Given a character $\chi : G_1 \times G_2 \to \mathbb{G}_m$, we get another character by

$$\chi \circ C_1 : G_1 \to \mathbb{G}_m, \quad \chi \circ C_2 : G_2 \to \mathbb{G}_m.$$

This defines a map

$$X^*(G_1 \times G_2) \ni \chi \mapsto (\chi \circ C_1, \chi \circ C_2) \in X^*(G_1) \times X^*(G_2).$$

Let $\lambda \in X^*(G_1)$, and $\mu \in X^*(G_2)$, then

$$(\lambda, \mu)(g_1, g_2) = \lambda(g_1)\mu(g_2),$$

this defines a map from $X^*(G_1) \times X^*(G_2)$ to $X^*(G_1 \times G_2)$. The maps

$$[\chi \mapsto (\chi \circ C_1, \chi \circ C_2)], [(\lambda, \mu) \mapsto [(g_1, g_2) \mapsto \lambda(g_1)\mu(g_2)]]$$

are inverse to one another. □

Definition 1.37. *An algebraic grup T is a torus if $T \cong \mathbb{G}_m^n$ for some $n \geq 0$.*

Corollary 1.4. *Let T bea torus then T is linearly reductive and we have a bijection*

$$X^*(T) \ni \chi \mapsto k_\chi \in \mathrm{Irr}(T).$$

Proof. By Lemma 1.9 and $\mathbb{G}_m$ is linearly reductive, T is linearly reducitve. Similarly by Lemma 1.8, we have

$$\mathrm{Irr}(T) = \mathrm{Irr}(\mathbb{G}_m)^n.$$

□

Remark 1.12. *Since each irreducible representations of $\mathbb{G}_m$ are exactly its character. We have*

$$X^*(T) = X^*(\mathbb{G}_m)^n.$$

In other words, any T-representation V has a canonical decomposition

$$V = \bigoplus_{\lambda \in X^*(T)} V_\lambda,$$

where

$$V_\lambda = \{v \in V \mid \forall t \in T, tv = \lambda(t)v\}.$$

Theorem 1.7. *Let $k = \mathbb{C}$, and $G \subset \mathrm{GL}_n$ be a closed subgroup.*

It is linearly reductive if it is stable under the involution $A \mapsto A^* := \overline{A}^T$.

Let V be an irreducible G-representation. Then for $N >> 0$, $V \otimes \det^{\otimes N}$ is a direct summand of $V^{\otimes r}$ for some $r \geq 0$ where det *is the one dimensional G-representation,*

$$G \hookrightarrow \mathrm{GL}_n \xrightarrow{\det} \mathbb{C},$$

$V = k^n$ is the standard GL_n-representation (the action is the linear transform by a matrix).

Proof. (Foundation course). □

Theorem 1.8 (Peter-Weyl)**.**
Let G be a linearly reductive algebrai group. Then we have

$$\mathcal{O}(G) \cong \bigoplus_{V \operatorname{Irr}(G)} V^* \boxtimes V,$$

where the inclusion $V^* \boxtimes V \hookrightarrow \mathcal{O}(G)$ is given by,

$$f \otimes v \mapsto (g \mapsto f(g^{-1}v)).$$

Proof. Foundation course. □

Example 1.13. *Let $G = \mathbb{G}_m = \mathrm{GL}_1$ and the action of $G \times G$ on $\mathcal{O}(G) = k[t, t^{-1}]$ is given by*

$$(t_1, t_2) \cdot T^n = t_1^{-n} t_2^n T^n.$$

Then $k \cdot T^n$ is isomorphic to $k_{-n} \boxtimes k_n \cong k_n^* \boxtimes k_n$ as $\mathbb{G}_m \times \mathbb{G}_m$-representation. Therefore

$$\mathcal{O}(\mathbb{G}_m) = \bigoplus_{n \in \mathbb{Z}} k \cdot T^n \cong \bigoplus n \in \mathbb{Z} k_n^* \boxtimes k_n.$$

Example 1.14. *Let G be an affine algebraic group and V be a finite dimensional G-representation. Define a function,*

$$\chi_V : G \to k, g \mapsto \operatorname{tr}(g|_V).$$

Explicity, this is the composition

$$G \xrightarrow{\rho} \mathrm{GL}(V) \xrightarrow{\operatorname{tr}} k$$

This lies in $\mathcal{O}^G$, where G acts on $\mathcal{O}(G)$ via

$$(gf)(x) = f(g^{-1}xg).$$

Example 1.15. *Let $V = k^n$ be the standard GL_n-representation. Then we define*

$$\chi_{\det}(g) = \deg(g).$$

Lemma 1.11. *Let U, U_1, U_2 be G-representations. We have*

1). $\chi_{U_1 \oplus U_2} = \chi_{U_1} + \chi_{U_2}$

2). $\chi_{U_1 \otimes U_2} = \chi_{U_1} \chi_{U_2}$

Let $C : G \to G$ be such that $i(g) = g^{-1}$ then

$$\chi_{U^*} = \chi_U \circ i.$$

If G is a finite group and $k = \mathbb{C}$, then

$$\chi_{U^*}(g) = \overline{\chi_U(g)}.$$

Proof. First two statements are assigned as exercises.

If G is finite then we have

$$g^{|G|} = e. \tag{1.8.1}$$

Under the representation, we get $\pi(g)$ is a matrix whose minimal polynomial divides $T^{\mathrm{ord}(g)} - 1$. In particular, $\pi(g)$ has distinct eigenvalues. In other words, it is diagonalizable.

$$\pi(g) \sim \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

For each i we have $\lambda_i^{\mathrm{ord}(g)} = 1$. Therefore,

$$\lambda_i^{-1} = \overline{\lambda}_i. \tag{1.8.2}$$

Thus we obtain

$$\chi_{U^*}(g) = \chi(g^{-1}) = \mathrm{tr}(g^{-1}) = \sum_{i=1}^{n} \lambda_i^{-1} = \overline{\sum_{i=1}^{n} \lambda_i} = \overline{\chi_U(g)}.$$

□

Theorem 1.9. *Let G be a linearly reductive group. Then the characters of the irreducible G-representations form a basis of $\mathcal{O}(G)^G$.*

Proof. By Theorem 1.8, we have

$$\mathcal{O}(G)^G = \left(\bigoplus_{U \in \mathrm{Irr}(G)} U^* \boxtimes U \right)^G \cong \bigoplus_{U \in \mathrm{Irr}(G)} \mathrm{Hom}_G(U, U).$$

By Schur's lemma, we get

$$\mathcal{O}(G)^G \cong \bigoplus_{U \in \mathrm{Irr}(G)} \mathrm{Hom}_G(U, U) = k \,\mathrm{id}_n \,.$$

Let $\{u_i\}_{i=1,\cdots,n}$ be a basis of U, then

$$tr(g|_U) = \chi_U(g) = (g \mapsto \sum_{i=1}^{n} u_i^*(g \cdot u_i)) \leftarrow\!\shortmid \sum_{i=1}^{n} u_i^* \boxtimes u_i \leftarrow\!\shortmid u \,\mathrm{id}_u \,.$$

We conclude χ_U forms a basis of $\mathcal{O}(G)^G$. □

Corollary 1.5. *Let V, V' be representation of a linearly reductive group G. Then we have*

$$V \cong V' \Leftrightarrow \chi_U = \chi_{U'}.$$

Proof. Let $V = \bigoplus_{U \in \mathrm{Irr}(G)} U^{\oplus d_U}$, and $V' = \bigoplus_{U \in \mathrm{Irr}(G)} U^{\oplus d'_U}$. Then

$$\chi_V = \sum_{U \in \mathrm{Irr}(G)} d_U \cdot \chi_U, \quad \chi_{V'} = \sum_{U \in \mathrm{Irr}(G)} d'_U \cdot \chi_U,$$

By the previous theorem, we get

$$\chi_V = \chi_{V'} \Leftrightarrow \forall U \in \mathrm{Irr}(G), d_U = d'_U \Leftrightarrow V \cong V'.$$

□

Let G be a finite group and $k = \mathbb{C}$, Note that G is linearly reductive as we have $G \hookrightarrow \mathfrak{S}_N \hookrightarrow \mathrm{GL}_N$. Hence we get

$$|\mathrm{Irr}(G)| = \dim \mathcal{O}(G)^G = \dim\{\theta : X^*(G) \to \mathbb{C}\} = |X^*(G)|.$$

Example 1.16. *Let $G = \mathfrak{S}_3$, then it has the following irreducible representations.*

$$\mathrm{triv}, \mathrm{sgn}, \nu = \mathbb{C}^3/\mathrm{triv},$$

Where $V_\nu = \{(x_1, x_2, x_3) \mid x_1 + x_2 + x_3 = 0\}$. And $\chi_{\mathbb{C}^3}$ is a trace. We have the following table,

	χ_{triv}	χ_{sgn}	χ_ν
e	1	1	2
(12)	1	−1	0
(123)	1	1	−1

Explicitly we have

$$\chi_\nu = \chi_{\mathbb{C}^3} - \chi_{\mathrm{triv}}$$

Let G be finite and $k = \mathbb{C}$, we can define a hermitian inner product on $\mathcal{O}(G)^G$ by

$$\langle b_1, b_2 \rangle = \frac{1}{|G|} \sum_{g \in G} b_1(g)\overline{b_2(g)}.$$

Lemma 1.12. *If G is finite and $k = \mathbb{C}$, Let V be a finite dimensional representation. Then we have*

$$\dim V^G = \langle \chi_\nu, \chi_{\mathrm{triv}} \rangle.$$

Proof. Let $\pi : V \to V$ be such that

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} g \cdot v.$$

Observe that $\pi|_{V^G} = \mathrm{id}_{V^G}$ and $\mathrm{im}\,\pi \subseteq V^G$. In other words, π is a projection onto V^G(with complement $\sum_{\substack{W \subseteq V \text{ irreducible.} \\ W \neq \mathrm{triv}}} W$). Thus

$$\dim V^G = \mathrm{tr}(\pi) = \frac{1}{|G|} \sum_{g \in G} \mathrm{tr}(g|_V) = \langle \chi_\nu, \chi_{\mathrm{triv}} \rangle.$$

The first equality follows from that if $V = W_1 \oplus W_2$ and $\rho : V \twoheadrightarrow W_1 \hookrightarrow V$ then $\mathrm{tr}(\rho) = \dim W_1$. □

Proposition 1.9. *Let G be a finite group and $k = \mathbb{C}$. If $U_1, U_2 \in \mathrm{Irr}(G)$, then*

$$\langle \chi_{U_1}, \chi_{U_2} \rangle = \begin{cases} 1 & (U_1 \cong U_2) \\ 0 & (U_1 \ncong U_2) \end{cases}.$$

In other words, the characters of the irreducible G-representations form an orthonormal basis of $\mathcal{O}(G)^G$.

Proof. By Schur's lemma, we know

$$\mathrm{Hom}_G(U_1, U_2) = \begin{cases} 1 & (U_1 \cong U_2) \\ 0 & (U_1 \ncong U_2) \end{cases}.$$

Note that $b_1, b_2, b_3 \in \mathcal{O}(G)^G$, we have

$$\langle b_1 b_2, b_3 \rangle = \langle b_1, \overline{b_2} b_3 \rangle.$$

By using the previous lemma, we get

$$\begin{aligned} \dim \mathrm{Hom}_G(U_1, U_2) &= (U_2 \underset{\mathbb{C}}{\otimes} U_1^*)^G, \\ &= \langle \chi_{U_2 \otimes U_1^*}, \chi_{\mathrm{triv}} \rangle, \\ &= \langle \chi_{U_2} \chi_{U_1^*}, \chi_{\mathrm{triv}} \rangle, \\ &= \langle \chi_{U_2}, \chi_{U_1} \rangle. \end{aligned}$$

□

Theorem 1.10. *Let $G = \mathrm{GL}_n$ and $k = \mathbb{C}$. Let*

$$T := \left\{ \begin{pmatrix} * & 0 & \cdots & 0 \\ 0 & * & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & * \end{pmatrix} \right\} subseteq \mathrm{GL}_n$$

be a torus consisting of diagonal matrices. By restricting along $T \hookrightarrow \mathrm{GL}_n$, we get

$$\mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n} \hookrightarrow \mathcal{O}(T)^{\mathfrak{S}_n},$$

where $\mathfrak{S}_n$ acts on T by permuting the diagonal entries.

Proof. We will first show that

$$\varphi : \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n} \to \mathcal{O}(T)$$

is injective.

Assume $\varphi(b) = 0$ but $b \not\equiv 0$. Let $A \in \mathrm{GL}_n$ be such that $b(A) \neq 0$.

Since b is GL_n invariant and any element of GL_n is conjugate toan upper triangular matrix, we may assume that A is also upper triangular such that

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Let

$$D_t = \begin{pmatrix} t^n & 0 & \cdots & 0 \\ 0 & t^{n-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & t \end{pmatrix}.$$

Then $0 \neq b(A) = b(D_t A D_t^{-1})$. Note that

$$D_t A D_t^{-1} = \begin{pmatrix} a_{11} & a_{12}t & \cdots & a_{1n}t^{n-1} \\ 0 & a_{22} & \cdots & a_{2n}t^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Taking $t \to 0$, we get

$$\lim_{t\to 0} D_t A D_t^{-1} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Since. $b \in \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}$ is a polynomial function, hence b is continuous in t. Therefore

$$0 \neq b(A).$$

In particular, we get

$$b\left(\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}\right) = b(\lim_{t\to 0} D_t A D_t^{-1}) = \lim_{t\to 0} b(D_t A D_t^{-1}) = \lim_{t\to 0} b(A) = b(A) \neq 0.$$

This contradicts to the choice of b, more explicitly,

$$0 = \varphi(b) = b|_T.$$

Thus we have proven the injectivity $\mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n} \hookrightarrow \mathcal{O}(T)$.

Since for any $D_1, D_2 \in T$, if $D_1 \sim D_2$ in T so are in GL_n. In particular, $D \in T$ is fixed by GL_n then so by $\mathfrak{S}_n$. Thus we have

$$\operatorname{im} \varphi \subseteq \mathcal{O}(T)^{\mathfrak{S}_n}.$$

□

Corollary 1.6. *Let $k = \mathbb{C}$ and U_1, U_2 be GL_n-representations. They are isomorphic as GL_n-representation if and only if they are isomorphic as T-representation.*

Proof. Any representation ofa linearly reductive group (such as $\mathrm{GL}_n(\mathbb{C})$ or T) is uniquely determined by its charactes. We have

$$U_1 \overset{\mathrm{GL}_n}{\cong} U_2 \Rightarrow U_1 \overset{T}{\cong} U_2$$

On the other hand,

$$U_1 \overset{T}{\cong} U_2 \Rightarrow \chi_{U_1}|_T = \chi_{U_2}|_T \Rightarrow \chi_{U_1} = \chi_{U_2} \Rightarrow U_1 \overset{\mathrm{GL}_n}{\cong} U_2.$$

□

2 Symmetric Polynomials

Remark 2.1. *We have*

$$\mathcal{O}(T) = \mathbb{C}[T_1^{\pm 1}, \cdots, T_n^{\pm 1}].$$

With this, we observe that

$$\mathcal{O}(T)^{\mathfrak{S}_n} = \mathbb{C}[T_1^{\pm 1}, \cdots, T_n^{\pm 1}]^{\mathfrak{S}_n} = \mathbb{C}[T_1, \cdots, T_n][(T_1 \cdots T_n)^{-1}].$$

For the last equality, observe that $T_i = (T_1 \cdots T_n)^{-1}(T - 1 \cdots T_{i-1}T_{i+1} \cdots T_n)$.

Notation 2.1.

$$\Lambda_n := \mathbb{Z}[T_1, \cdots, T_n]^{\mathfrak{S}_n}, \quad \Lambda_n^r = \{p \in \Lambda_n \mid p \text{ is homogeneous of degree } r\}.$$

Remark 2.2. Λ_n *has a graded algebra by*

$$\Lambda_n = \bigoplus_{r \in \mathbb{N}} \Lambda_n^r.$$

Definition 2.1. *A decreasing sequence (ie. for any* $i \in \mathbb{N}$, $\lambda_i \geq \lambda_{i+1}$*) of non-negative integers* $\lambda = (\lambda_i)_{i \in \mathbb{N}}$ *is called a partition.*

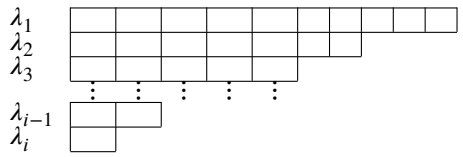
We call λ *is partition of* n *if* $n = \sum_{i \in \mathbb{N}} \lambda_i$.

If $i + 1$ *is minimal index such that* $\lambda_{i+1} = 0$*, then we say* λ *has* i*-parts.*

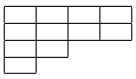
Notation 2.2. *For a given partition* λ*, we write* $\lambda = (1^{\alpha_1} 2^{\alpha_2} \cdots)$ *where* α_i *denotes the number of times the number* i *appears in the term of the sequence.*

Example 2.1. *For* $(4, 4, 2, 1)$*, we write* (124^2).

Definition 2.2. *For a partition* λ *which has* i *parts, the Young's diagram of* λ *is a diagram such that*



Example 2.2. *For* $(4, 4, 2, 1)$*, its Young diagram is*



Definition 2.3. *For any* $k \geq 0$*, the power sum is*

$$\psi_k = \sum_{i=1}^{n} T_n^k.$$

This is clearly a symmetric polynomial.

Definition 2.4. *The elementary symmetric polynomials are defined by expressing products*

$$\prod_{i=1}^{n}(1 - T_i x) \in \Lambda_n[x]$$

in terms of powers of x,

$$\prod_{i=1}^{n}(1 - T_i x)1 + \sum_{i=1}^{n}(-1)^i e_i x^i,$$

where $e_i \in \Lambda_n$. More explicitly,

$$e_i = \sum_{1 \le a_1 \le \cdots \le a_i \le n} T_{a_1} \cdots T_{a_i}.$$

Example 2.3.

$$e_1 = \sum_{i=1}^{n} T_i, \quad e_n = T_1 \cdots T_n.$$

Definition 2.5. *Let λ be a partition of n which has at most n-parts. Then we define*

$$m_\lambda = \sum_{\sigma \in \mathfrak{S}_n} \sigma \prod_{i=1}^{n} T_i^{\lambda_i}.$$

In other words, m_λ is the orbit of $\prod_{i=1}^{n} T_i^{\lambda_i}$ under $\mathfrak{S}_n$. We call m_λ a monomial symmetric polynomial.

Remark 2.3. *The above definition generalize the symmetric polynomials we have defined. Explicitly,*

$$\psi_i = m_{(i)}, \quad e_i = m_(111 \cdots 10 \cdots 0)$$

where for e_i, there are 1-s in the first i-terms.

Remark 2.4. *Any monomial of Λ_n is a $\mathfrak{S}_n$ orbit of some mononial of the form,*

$$T_1^{\lambda_1} \cdots T_n^{\lambda_n}$$

where λ is a partition of r which has at most n-parts. Indeed, this forms a basis of Λ_n^r.

Moreover, we have

$$\{m_\lambda \mid \lambda \text{ has at most n-parts}\},$$

is a basis of the algebra Λ_n.

Definition 2.6. *Given a partition $\lambda = (1^{k_1}, \cdots, n^{k_n}, \cdots)$, we define*

1. $a(\lambda) = k_1! \cdots k_n! \cdots$,
2. $b(\lambda) = \prod_{i \in \mathbb{N}} i^{k_i}$,
3. $n(\lambda) = a(\lambda)b(\lambda)$.

Definition 2.7.

$$\beta_\lambda(\mu) := \chi_{\mathbb{C}[S_r/S_\lambda]}(\mu).$$

Proposition 2.1. *Let $\lambda, \mu \vdash r$ with $\lambda = (\lambda_1 > \cdots > \lambda_k)$ and $\mu = (1^{p_1} \cdots n^{p_n})$. Then*

$$\beta_\lambda(\mu) = \sum_{\substack{\mu = \bigoplus_{i=1}^k \mu_i \\ \mu_i \vdash \lambda}} \prod_{j=1}^{n} \binom{p_j}{p_{j1}, p_{j2} \cdots p_{jk}}.$$

where $\mu_i = (1^{p_{1i}} 2^{p_{2i}} \cdots)$

Proof.

$$\beta_\lambda(\mu) = \sum_i \frac{|S_r(\mu)|}{|S_\lambda(h_i)|},$$

where h_i are representatives of the S_λ conjugacy classes in $C_{S_r}(\mu)$ (ie a conjugacy class is S_r of cycle type μ). Not $|S_r(\mu)| = |S_r(\sigma)|$ for any σ of cycle type μ.

We have

$$C_{S_{\lambda_1}}(\mu_1) \times \cdots \times C_{S_{\lambda_k}}(\mu_k) \subseteq C_{S_r}(\mu).$$

Thus we have

$$(\mu_i \vdash \lambda_i) \Leftrightarrow \mu = \bigoplus_{i=1}^{k} \mu_i.$$

Now $|S_r(\mu)| = n(\mu)$ from the lemma, using this, we obtain,

$$|S_{\lambda_1}(\mu_1) \times \cdots \times S_{\lambda_k}(\mu_k)| = \prod_{i=1}^{k} n(\mu_i).$$

For $\mu = \bigoplus_{i=1}^k \mu_i$, we get

$$b(\mu) = \prod_{i=1}^{k} b(\mu_i).$$

For $a(\mu)$, we have,

$$a(\mu) = p_1! p_2! \cdots, \quad a(\mu_i) = (p_{1i})!(p_{2i})! \cdots.$$

Thus we obtain,

$$\prod_{i=1}^{n} a(\mu_i) = \prod_{i=1}^{n} \prod_{j=1}^{k} (p_{ij})!.$$

Substituting these, we get

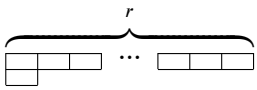
$$\beta_\lambda(\mu) = \sum_{\substack{\mu_i = \bigoplus_{i=1}^k \mu_i \\ \mu \vdash \lambda}} \frac{a(\mu) b(\mu)}{\prod_{i=1}^n a(\mu_i) b(\mu_i)} = \sum_{\substack{\mu = \bigoplus_{i=1}^k \mu_i \\ \mu_i \vdash \lambda}} \prod_{j=1}^{n} \frac{p_j!}{(p_{j!})! \cdots (p_{jk})!}.$$

Using the generalized binomial coefficient, we get

$$\beta_\lambda(\mu) = \sum_{\substack{\mu_i = \bigoplus_{i=1}^k \mu_i \\ \mu \vdash \lambda}} \prod_{j=1}^{n} \binom{p_j}{p_{j1}, \cdots, p_{jk}}.$$

□

Example 2.4. *Let $V = \mathbb{C}^r = C[S_r/S_\lambda]$ where $\lambda = (r-1 \geq 1)$.*

$\lambda \sim$ 

For any $\mu = (1^{p_1} 2^{p_2} \cdots)$, there is at most one way to write μ as the direct sum of partitions,

$$\mu = \mu_1 \oplus \mu_2$$

with $\mu_1 \vdash \lambda_1 = r-1, \mu_2 \vdash \lambda_2 = 1$. Namely,

$$\mu_2 = (1), \mu_1 = (1^{p_1 - 1} 2^{p_2} 3^{p_3} \cdots).$$

Notice this is only possible if and only if $p_1 > 0$. Then using the proposition,

$$\beta_\lambda(\mu) = \binom{p_1}{p_1 - 1, 1}\binom{p_2}{p_2, 0}\binom{p_3}{p_3, 0} = p_1.$$

Recall for λ a partition with at most n-parts, $m_\lambda \in \Lambda_n$ is the sum of all monomials in the S_n-orbit of $T_1^{\lambda_1} T_2^{\lambda_2} \cdots$. We also have defined that

$$\psi_k = \sum_{i=1}^{n} T_i^k \in \Lambda_n.$$

For any partition $\mu = (\mu_1 \geq \mu_2 \geq \cdots)$, define

$$\psi_\mu = \psi_{\mu_1} \psi_{\mu_2} \cdots.$$

Lemma 2.1. *We have,*

$$\psi_\mu = \sum_{\substack{\lambda \\ \lambda \text{ has at most } n\text{-parts.}}} \beta_\lambda(\mu) m_\lambda,$$

for all $\mu \vdash r$.

Proof. Note $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \lambda_k)$ and $\mu = (1^{p_1} 2^{p_2} \cdots)$. Then

$$\psi_\mu = (\psi_1)^{p_1} (\psi_2)^{p_2} \cdots = \sum_{\{1, \cdots, p_j\} = S_{j1} \coprod \cdots \coprod S_{jk}} \prod_{i,j} T_i^{j|S_{ji}|}.$$

Here we multiply, out $\psi_j^{p_j}$ by picking T_i^j for each index in S_j. This means that the coefficient of $T_1^{\lambda_1} T_2^{\lambda_2} \cdots T_k^{\lambda_k}$ in ψ_μ is equal to all possible ways to partition $\{1, \cdots, p_j\} = S_{j1} \coprod \cdots \coprod S_{jk}$. Such that

$$\lambda_i = p_{1i} + 2p_{2i} + \cdots \text{ where } p_{ji} = |S_{ji}| \tag{2.0.1}$$

Set $\mu_i := (1^{p_{1i}} 2^{p_{2i}} \cdots)$, then $\mu_i \vdash \lambda$ if and only if Equation (2.0.1) holds. Also $\mu = \bigoplus_{i=1}^k \mu_i$ since $p_j = \sum_{i=1}^k p_{ji}$.

For a fixed decomposition $\mu = \bigoplus_{i=1}^k \mu_i$, the number of ways to choose $S_{j1} \sqcup S_{j2} \sqcup \cdots \sqcup S_{jk} = \{1, \cdots, p_j\}$, with $\S_{ji} = p_{ji}$ is given by

$$\binom{p_j}{p_{j1}, \cdots, p_{jk}}$$

Therefore, coefficient of $T_1^{\lambda_1} \cdots T_k^{\lambda_k}$ in ψ_μ is

$$\sum_{\substack{\mu = \bigoplus_{i=1}^k \mu_i \\ \mu_i \vdash \lambda}} \prod_{j=1}^n \binom{p_j}{p_{j1}, \cdots, p_{jk}} = \beta_\lambda(\mu).$$

Note m_λ is the unique monomial symmetric polynomial containing $T_1^{\lambda_1} \cdots T_{\lambda_k}$, thus

$$\psi_\mu = \sum_\lambda \beta_\lambda(\mu) m_\lambda.$$

□

We can also express ψ_μ in the Schur basis,

$$\psi_\mu = \sum_\lambda c_\lambda(\mu) s_\lambda, \tag{2.0.2}$$

where $c_\lambda \in \mathbb{Z}$. Note that we can view $c_\lambda \in \mathcal{O}(S_r)^{S_r}$, where $c_\lambda(\sigma) = c_\lambda(\mu)$ if σ has cycle of type μ.

Proposition 2.2. *Let $\lambda, \mu \vdash r$ with at most n-parts. Then*

$$\langle c_\lambda, c_\mu \rangle = \delta_{\lambda,\mu},$$

(with respect to the standard Hermitian product $\langle f, g \rangle = \frac{1}{|S_r|} \sum_{\sigma \in S_r} f(\sigma)\overline{g(\sigma)}$). We see the c_λ-s are orthonormal.

Proof. Recall the power series in $\mathbb{C}[\![x]\!]$. We can define,

$$\ln(1+x) = \sum_{n \in \mathbb{N}} (-1)^{n-1} \frac{x^n}{n}, \quad \exp(x) = \sum_{n \in \mathbb{N}_0} \frac{x^n}{n!}.$$

They satisfy the following properties which can be verified by simple computations.

1). $\ln((1+x)(1+y)) = \ln(1+x) + \ln(1+y)$,

2). $\exp(\ln(1+x)) = 1+x$.

We use Cauchy's formula, such that

$$\prod_{1\le i,j\le n} \frac{1}{1-x_i y_j} = \sum_{\lambda} s_\lambda(x_1,\cdots,x_n) s_\lambda(y_1,\cdots,y_n).$$

Applying ln to this, we get,

$$\begin{aligned}
\ln\left(\sum_{\lambda} s_\lambda(x_1,\cdots,x_n) s_\lambda(y_1,\cdots,y_n)\right) &= \ln\left(\prod_{1\le i,j\le n} \frac{1}{1-x_i y_j}\right), \\
&= \sum_{1\le i,j\le n} \ln\left(\frac{1}{1-x_i y_j}\right), \\
&= -\sum_{1\le i,j\le n} \ln\left(1-x_i y_j\right), \\
&= -\sum_{1\le i,j\le n} \sum_{k\in\mathbb{N}} (-1)^{k-1} \frac{(-x_i y_j)^k}{k}, \\
&= \sum_{1\le i,j\le n} \sum_{k\in\mathbb{N}} \frac{(x_i y_j)}{k}, \\
&= \sum_{k\in\mathbb{N}} \frac{1}{k} \sum_{1\le i,j\le n} (x_i y_j)^k, \\
&= \sum_{k\in\mathbb{N}} \frac{\psi_k(x_1,\cdots,x_n)\psi_k(y_1,\cdots,y_n)}{k}.
\end{aligned}$$

Apply exp to this, we obtain,

$$\sum_{\lambda} s_\lambda(x_1,\cdots,x_n) s_\lambda(y_1,\cdots,y_n) = \exp\left(\sum_{k\in\mathbb{N}} \frac{\psi_k(x_1,\cdots,x_n)\psi_k(y_1,\cdots,y_n)}{k}\right).$$

By expanding x, we obtain,

$$= \sum_{k\in\mathbb{N}_0} \frac{1}{k!} \left(\sum_{h\in\mathbb{N}} \frac{\psi_h(x_1,\cdots,x_n)\psi_h(y_1,\cdots,y_n)}{h}\right)^k,$$

$$= \sum_{k\in\mathbb{N}_0} \frac{1}{k!} \sum_{k=k_1+k_2\cdots} \binom{k}{k_1,k_2,\cdots} \left(\frac{\psi_1(x_1,\cdots,x_n)\psi_1(y_1,\cdots,y_n)}{1}\right)^{k_1} \left(\frac{\psi_2(x_,\cdots,x_n)\psi_2(y_1,\cdots,y_n)}{1}\right)^{k_2} \cdots$$

Simplyfy this, we get

$$= \sum_{k\in\mathbb{N}_0} \sum_{k=k_1+k_2+\cdots} \frac{1}{(k_1!k_2!\cdots)(1^{k_1}2^{k_2}\cdots)} (\psi_1(x_1,\cdots,x_n)^{k_1}\psi_2(x_1,\cdots,x_n)^{k_2}\cdots)(\psi_1(y_1,\cdots,y_n)^{k_1}\psi_2(y_1,\cdots,y_n)^{k_2}\cdots),$$

$$= \sum_{\mu} \frac{1}{n(\mu)} \psi_\mu(x_1,\cdots,x_n)\psi_\mu(y_1,\cdots,y_n),$$

where $\mu = (1^{k_1}2^{k_2}\cdots)$. Substitute $\psi_\mu = \sum_\lambda c_\lambda(\mu)s_\lambda$ into the above equation, we get,

$$\begin{aligned}
&= \sum_\mu \frac{1}{n(\mu)} \left(\sum_{\lambda_1} c_{\lambda_1}(\mu) s_{\lambda_1}(x_1, \cdots, x_n) \right) \left(\sum_{\lambda_2} c_{\lambda_2}(\mu) s_{\lambda_2}(y_1, \cdots, y_n) \right), \\
&= \sum_{\lambda_1} \sum_{\lambda_1} \left(\sum_\mu \frac{1}{n(\mu)} c_{\lambda_1}(\mu) c_{\lambda_2}(\mu) \right) s_{\lambda_1}(x_1, \cdots, x_n) s_{\lambda_2}(y_1, \cdots, y_n), \\
&= \sum_\lambda s_\lambda(x_1, \cdots, x_n) s_\lambda(y_1, \cdots, y_n).
\end{aligned}$$

The $s_{\lambda_1}(x_1, \cdots, x_n)s_{\lambda_2}(y_1, \cdots, y_n)$ are linearly independent. We get

$$\sum_\mu \frac{1}{n(\mu)} c_{\lambda_1}(\mu) c_{\lambda_2}(\mu) = \delta_{\lambda_1, \lambda_2}.$$

where $\frac{|S_r|}{n(\mu)}$ denotes the number of conjugacy classes of type μ. We can also write this as

$$= \frac{1}{S_r} \sum_\mu \frac{|S_r|}{n(\mu)} c_{\lambda_1}(\mu) c_{\lambda_2}(\mu) = \frac{1}{|S_r|} \sum_{\sigma \in S_r} c_{\lambda_1}(\mu) c_{\lambda_2}(\mu).$$

By definition $c_{\lambda_2}(\mu)$ is an integer, we get

$$\frac{1}{|S_r|} \sum_{\sigma \in S_r} c_{\lambda_1}(\mu) \overline{c_{\lambda_2}(\mu)} = \langle c_{\lambda_1}, c_{\lambda_2} \rangle.$$

□

Theorem 2.1. *Let $\chi_\lambda = \chi_{S^\lambda}$ be the character ofthe Specht module S^λ. We have the following statements.*

$\forall \lambda, \lambda \vdash r$ and λ has at most n-parts, we have $\chi_\lambda = c_\lambda$. In particular, the c_λ-s are irreducible characters. If $n \geq r$, then c_λ are all the irreducible characters.

Furthermore, we have,

$$\beta_\lambda = \sum_{\phi \geq \lambda} k_{\phi,\lambda} c_\phi.$$

Proof. Recall

i). $\psi_\mu = \sum_\lambda c_\lambda(\mu)s_\lambda$,

ii). $m_\phi = \sum_{\phi \geq \lambda} p_{\phi,\lambda} s_\lambda$.

iii). $s_\phi = \sum_{\phi \geq \lambda} k_{\phi,\lambda} m_\lambda$.

We have,

$$\sum_\lambda c_\lambda(\mu)s_\lambda = \psi_\mu = \sum_\phi \beta_\phi(\mu) m_\phi = \sum_\phi \sum_{\phi \geq \lambda} \beta_\phi(\mu) p_{\phi,\lambda} s_\lambda.$$

By comparing the coefficients, which form a unitriangular matrix,

$$c_\lambda(\mu) = \sum_{\phi \geq \lambda} \beta_\phi p_{\phi,\lambda} = \beta_\lambda + \sum_{\phi > \lambda} p_{\phi,\lambda} \beta_\phi.$$

The $(k_{\phi,\lambda})$ are the inverse matrix of $(p_{\phi,\lambda})$, hence,

$$\beta_\lambda = c_\lambda + \sum_{\phi>\lambda} k_{\phi,\lambda} c_\lambda. \tag{2.1.1}$$

Note that β_λ is a character of the representation by definition. Thus β_λ is an integer linear combination of the χ_λ-s. Thus each c_λ is an integer linear combination of χ_μ. In another words,

$$c_\lambda = \sum r_\mu \chi_\mu, \quad r_\mu \in \mathbb{Z}.$$

With this expression, we get ,

$$1 = \langle c_\lambda, c_\lambda \rangle = \sum_\mu r_\mu^2 \langle \chi_\mu, \chi_\mu \rangle = \sum_\mu r_\mu^2.$$

Since each r_μ is an integer, we know that they are all 0 except for one $\mu = \lambda'$, and $r_{\lambda'} = \pm 1$. This means, for some λ', we get,

$$c_\lambda = \pm \chi_{\lambda'}.$$

If the coefficient is -1, then,

$$\beta_\lambda = -\chi_{\lambda'} + \sum_{\phi>\lambda} k_{\phi,\lambda} \chi_{\phi'},$$

which is a contradiction, since β_λ is a non-negative integer linear combination of the χ_ϕ's. Thus we obtain, for some λ',

$$c_\lambda = \chi_{\lambda'}.$$

Remains to show that $\lambda = \lambda'$. We prove this by downward induction on the lexicographical order.

If we take $\lambda = (r)$, then $s^\lambda = \mathrm{triv}$. We also have,

$$c\lambda = \beta_\lambda = \chi_{\mathbb{C}[S_r/S_\lambda]} = \chi_{\mathrm{triv}}.$$

Suppose $c_\phi = \chi_\phi$ for all $\phi > \lambda$. Then substituting this in Equation (2.1.1), we get

$$\begin{aligned} \beta_\lambda &= c_\lambda + \sum_{\phi>\lambda} k_{\phi,\lambda} c_\phi \\ &= \chi_{\lambda'} + \sum_{\phi>\lambda} k_{\phi,\lambda} \chi_\phi. \end{aligned}$$

By previous lemma, we have s^λ apppears in $\mathbb{C}[S_r/S_\lambda]$ with multiplicity 1. Thus χ_λ appears in β_λ, in particular

$$\Rightarrow \chi_\lambda = \chi_{\lambda'} \Rightarrow \lambda = \lambda'.$$

□

3 Schur-Weyl Duality

We assume $k = \mathbb{C}$. The goal of the section is to determine and construct all irreducible representations of GL_n and SL_n.

We have seen earlier the following.

1. Any irreducible GL_n-representation appears in some $V^{\otimes r} \otimes \det^{\otimes m}$ where $V = C^n$ with the natural vector representation.

2. Similarly, any irreducible representation of SL_n appears in $V^{\otimes r} \otimes \det^{\otimes m}$.

Remark 3.1. *$V^{\otimes r}$ are not irreducible in general.*

Example 3.1.

$$V \otimes V = S^2 V \otimes \bigwedge^2 V.$$

Remark 3.2. *S_r acts on $V^{\otimes r}$ by permuting tensor factors. Explicitly,*

$$\sigma(v_1 \otimes \cdots \otimes v_r) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(r)}.$$

Clearly, the S_r-action and the GL_n-action commute. Since S_r acts diagonally on $V^{\otimes r}$. Thus we can view $V^{\otimes r}$ as a $\mathrm{GL}_n \times S_r$-representation.

In particular, there is a functor,

$$S(V) : \mathrm{Rep}(S_r) \to \mathrm{Rep}(\mathrm{GL}_n), \quad \mathrm{Rep}(S_r) \ni M \mapsto \mathrm{Hom}_{S_r}(M, \overset{\mathrm{GL}_n}{\curvearrowright} V^{\otimes r}) \in \mathrm{Rep}(\mathrm{GL}_n),$$

where GL_n acts on $V^{\otimes r}$.

Remark 3.3. *We can also view $V^{\otimes r}$ as a right S_r-representation by*

$$(v_1 \otimes \cdots \otimes v_r)\sigma = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(r)}.$$

With this, $V^{\otimes r}$ is a (GL_n, S_r)-bimodule/birepresentation.

Proposition 3.1. *Any S_r-representation is self-dual in other words,*

$$S(V)(M) = \mathrm{Hom}_{S_r}(M, V^{\otimes r}) = (V^{\otimes r} \underset{\mathbb{C}}{\otimes} M^*)^{S_r} = V^{\otimes r} \underset{\mathbb{C}[S_r]}{\otimes} M.$$

Proof. Foundation course. □

Theorem 3.1 (Double Centralizer Theorem)**.** *Let W be a finite dimensional complex vector space and $A \subset \mathrm{End}_{\mathbb{C}}(W)$ be a semisimple subalgebra of the endomorphisms of W. We define*

$$B :== \mathrm{End}_A(W),$$

then we have the following statements.

1). B is also semisimple.

2). $\mathrm{End}_B(W) = A$.

3). $W \cong \bigoplus_{i\in I} U_i \boxtimes W_i$ *as a* $A \otimes B$*-modules where* $\{U_i\}_{i\in I}$ *are the simple* A*-modules and* $W_i = \mathrm{Hom}_A(U_i, \overset{B}{\curvearrowright} W)$ *for each* $i \in I$.

4). $\{W_i \mid i \in I\}$ *are the all simple* B*-modules.*

5). There is a bijection $\mathrm{Irr}(A) \overset{1:1}{\Leftrightarrow} \mathrm{Irr}(B), U_i \mapsto \mathrm{Hom}_A(U_i, W) = W_i$.

Proof. Since A is semisimple we have,

$$A \cong \bigoplus_{i\in I} \mathrm{End}(U_i),$$

which follows from Artin-Wedderborn theorem. For each $i \in I$, there is a morphism of $A \otimes B$-modules,

$$\overset{A}{\curvearrowright} U_i \boxtimes \mathrm{Hom}_A\,(U_i, W) \overset{\mathrm{ev}}{\to} W, \quad u \otimes f \mapsto f(u).$$

Note that $\mathrm{Hom}_A(U_i, W) = W_i$. Then we get a morphism of $A \otimes B$-modules,

$$\bigoplus_{i\in I} U_i \boxtimes W_i \overset{\phi}{\to} W.$$

First, we will show ϕ is surjective. Indeed, if W is semisimple as an A-module (A is semisimple). Then we have,

$$W \cong \bigoplus_{i\in I} U_i^{\oplus n_i}.$$

Then we have,

$$U_i \boxtimes \mathrm{Hom}(U_i, U_i^{\oplus n_i}) \to U_i^{\oplus n_i} \tag{3.1.1}$$

is surjective. To see this, let us denote $U_i^{(j)} \subseteq U_i^{\oplus n_i}$ to be the j-th copy then

$$U_i \xrightarrow{\sim} U_i^{(j)} \hookrightarrow U_i^{\oplus n_i}, \qquad U_i \xrightarrow{f_j} U_i^{\oplus n_i}$$

We see $\mathrm{im}(f_j) = U_i^{(j)}$. Thus $U_i^{(j)}$ lies in the image of (3.1.1).

To prove ϕ is injective. We see,

$$\begin{aligned} \dim W_i = \dim \mathrm{Hom}_A(U_i, W) = n_i. \\ \Rightarrow \dim \bigoplus_{i\in I} U_i \boxtimes W_i = \sum_{i\in I} \dim U_i \cdot n_i, \\ = \dim W. \end{aligned}$$

Thus ϕ is an injective, in particular an isomorphism.

$$\Rightarrow B = \mathrm{End}_A(W) = \mathrm{End}_A\left(\bigoplus_{i\in I} U_i \boxtimes W\right) = \bigoplus_{i\in I} \mathrm{End}(W_i).$$

Thus we conclude B is semisimple and the simple B-modules are the W_i-s with $W_i \neq \{0\}$.

Note $W_i = \{0\}$, then $\mathrm{Hom}_{S_r}(U_i, W) = \{0\}$. This means that U_i does not appear in W. Thus

$$\mathrm{End}(U_i) = \bigoplus_{i \in I} \mathrm{End}(U_j) = A$$

acts by 0 on W. Since $A \subseteq \mathrm{End}(W)$, the only element $a \in A$ acting by 0 is $a = 0$.

$$\mathrm{End}_B(W) = \mathrm{End}_B\left(\bigoplus_{i \in I} U_i \boxtimes W_i\right) = \bigoplus_{i \in I} \mathrm{End}_B(U_i \boxtimes W_i) = \bigoplus_{i \in I} \mathrm{End}(U_i) = A.$$

□

Lemma 3.1. *The image of* $\mathrm{GL}_n \to \mathrm{End}(V^{\otimes r})$ *coming from* $\mathrm{GL}_n \curvearrowright V^{\otimes r}$ *generates* $\mathrm{End}_{S_r}(V^{\otimes r})$.

Proof. Let $B' \subseteq \mathrm{End}(V^{\otimes r})$ be the subalgebra generated by the image. Clearly we have $B' \subseteq B \subseteq \mathrm{End}_{S_r}(V^{\otimes r})$. We want to show $B' = B$.

Consider the morphism of varieties for each $X \in \mathrm{End}(V^{\otimes r})$,

$$\mathbb{A}^1_k \xrightarrow{\phi_X} \mathrm{End}(V^{\otimes r}) \cong \mathbb{A}^{(nr)^2}_k, \quad t \mapsto (t\,\mathrm{id}_V + x) \otimes \cdots \otimes (t\,\mathrm{id}_V + x).$$

Note that

$$\det(t\,\mathrm{id}_V + X) = t^n + a_{n-1}t^{n-1} + \cdots + a_0,$$

which has at most n-distinct roots. Thus $t\,\mathrm{id}_V + X \in \mathrm{GL}_n$ for all but finitely many t, we have,

$$\phi_X(t) \in B'.$$

In another words, $\varphi_X^{-1}(B)$ is dense in $\mathbb{A}^1_k$ since closed subsets of $\mathbb{A}^1_k$ are $\mathbb{A}^1_k$ and finite sets.

But ϕ_X is continuous and $B' \subset \mathrm{End}(V^{\otimes r})$ is closed since B' is a subvectorspace. We see

$$\phi_X^{-1}(B') \text{ is dense and closed} \Rightarrow \phi_X^{-1}(B') = \mathbb{A}^1_k \Rightarrow \mathrm{im}(\phi_X) \subseteq B'.$$

In particular

$$\forall X \in \mathrm{End}(V), X \otimes \cdots \otimes X = \phi_X(0) \in B'.$$

Note that,

$$\mathrm{End}_{S_r}(V^{\otimes r}) = \mathrm{End}_{\mathbb{C}}(V^{\otimes r})^{S_r} = (\mathrm{End}_{\mathbb{C}}(V)^{\otimes r})^{S_r}.$$

From the exercise,

Exercise 3.1. *For any representation* W, $(W^{\otimes r})^{S_r}$ *is spanned by* $w^{\otimes r} = \overbrace{w \otimes \cdots \otimes w}^{r}$, *then* $\mathrm{End}_{S_r}(V^{\otimes r})$ *is spanned by all* $X \otimes \cdots \otimes X \in B'$ *for* $X \in \mathrm{End}(V)$.

We conclude $B' = B$. □

Notation 3.1. *For any* $\lambda \vdash r$,

$$S_\lambda(V) = S(V)(S^\lambda) = \mathrm{Hom}_{S_r}(S^\lambda, \overset{\mathrm{GL}_n}{\curvearrowright} V^{\otimes r}).$$

Corollary 3.1 (Schur-Weyl Duality)**.** *We have*

$$V^{\otimes r} = \bigoplus_{\lambda \vdash r} \mathcal{S}_\lambda(V) \boxtimes S^\lambda$$

as $\mathrm{GL}_n \times S_r$*-representation.*

Moreover,

1. *For any* $\lambda \vdash r$*,* $\mathcal{S}_\lambda$*is either* 0 *or a simple* GL_n*-representation.*
2. *If* $\mathcal{S}_\lambda(V), \mathcal{S}_\lambda(V) \neq \{0\}$*, then* $\mathcal{S}_\lambda(V) \cong \mathcal{S}_\mu(V) \Leftrightarrow \lambda = \mu$*.*

In other words, there is a bijection

$$\{M \in \mathrm{Irr}(S_r) \mid M \text{ appears in } V^{\otimes r}\} \overset{1:1}{\leftrightarrow} \{M' \in \mathrm{Irr}(\mathrm{GL}_n) \mid M' \text{ appears in } V^{\otimes r}\}$$
$$S^\lambda \mapsto \mathcal{S}_\lambda(V).$$

Proof. Let $A = \mathrm{im}(\mathbb{C}[S_r] \to \mathrm{End}(V^{\otimes r}))$. Since $\mathbb{C}[S_r]$ is semisimple A is also semisimple. Let $B = \mathrm{End}_{S_r}(V^{\otimes r})$.

$$V^{\otimes r} \cong \bigoplus_{\substack{i \in I \\ S^\lambda \text{ is a simple } A\text{-module}}} \mathrm{Hom}(S^\lambda, V^{\otimes r}) \boxtimes S^\lambda,$$

as $B \otimes A$-modules where the sum runs over all simple A-modules which is a some subset of $\{S^\lambda \mid \lambda \vdash r\}$.

By restriction, we also have

$$V^{\otimes} \cong \bigoplus_{\lambda \vdash r} \mathrm{Hom}(S^\lambda, V^{\otimes r}) \boxtimes S^\lambda.$$

as $\mathrm{GL}_n \times S_r$-representations. here we take the sum over all $\lambda \vdash r$.

This woks since

$$\mathrm{Irr}(A) = \{S^\lambda \mid S^\lambda \text{ appears in } V^{\otimes r} \text{ as an } S_r\text{-representation}\}.$$

If S^λ does not appear in $V^{\otimes r}$, then

$$\mathrm{Hom}_{S^r}(S^\lambda, V^{\otimes r}) = 0.$$

By Theorem 3.1, we also have,

$$\{S_\lambda(V) \mid S_\lambda(V) \neq \{0\}\},$$

are the simple B-modules. Since GL_n generates B from Lemma 3.1, they are also the simple GL_n-representations appearing in $V^{\otimes r}$. □

Corollary 3.2. *Any irreducible* GL_n*-representation is of the form*

$$\mathcal{S}_\lambda(V) \otimes \det^{\otimes m}$$

for some $m \in \mathbb{Z}$*.*

Proof. If M is a simple GL_n-representation, then M appears in some $V^{\otimes r} \otimes \det^{\otimes l}$.

Note if $N \in \mathrm{Irr}(\mathrm{GL}_n)$, then $N \otimes \det \in \mathrm{Irr}(|GL_n)$. Therefore, we can rephrase this as

$$M \otimes \det^{\otimes -l} \text{ appears in } V^{\otimes r}.$$

which means,

$$M \otimes \det^{\otimes -l} \cong S_\lambda(V)$$

for some $\lambda \vdash r$. Thus we obtain,

$$M \cong S_\lambda(V) \otimes \det^{\otimes l}.$$

□

Remains to see when we have,

$$S_\lambda(V) \otimes \det^{\otimes m} \overset{?}{\cong} S_\mu(V) \otimes \det^{\otimes l},$$

and

$$S_\lambda(V) = \{0\}.$$

Example 3.2. *Let $n = 1, V = \mathbb{C}, \mathrm{GL}_n = \mathbb{C}^\times$. Then S_r acts trivially on $V^{\otimes r} \cong \mathbb{C}$.*

$$\to S_\lambda(V) = \mathrm{Hom}_{S_r}(S^\lambda, V^{\otimes r}) = \{0\},$$

unless $\lambda = (r)$.

Lemma 3.2. *Let $b = \sum_{\sigma \in S_m} \mathrm{sgn}(\sigma)\sigma \in \mathbb{C}[S_m]$. Then for any S_m-representation W, we have,*

$$bW = \{w \in W \mid \forall \sigma \in S_m, \sigma w = \mathrm{sgn}(\sigma) w\}.$$

Proof. We have

$$\sigma b = \mathrm{sgn}(\sigma) b \Rightarrow bW \subseteq \{w \in W \mid \forall \sigma \in S_m, \sigma w = \mathrm{sgn}(\sigma) w\}.$$

Let $w \in \{w \in W \mid \forall \sigma \in S_m, \sigma w = \mathrm{sgn}(\sigma) w\}$, then

$$\frac{b}{m!} w = \frac{1}{m!} \sum_{\sigma \in S_m} \mathrm{sgn}(\sigma) \sigma w = \frac{1}{m!} \sum_{\sigma \in S_m} w = w.$$

Thus this completes the proof. □

Proposition 3.2. *we have $S_\lambda(V) \neq \{0\} \Leftrightarrow \lambda$ has at most n-parts.*

Proof. For any vectorspace W,

$$\wedge^m W = \{x \in W^{\otimes r} \mid \forall \sigma \in S_m, \sigma x = \mathrm{sgn}(\sigma) x\} \subset W^{\otimes r} \overset{\text{Lemma 3.2}}{=} bW^{\otimes m}. \quad (3.1.2)$$

Let $\lambda \vdash r$ with more than n-parts. Take T to be the column superstandard tableau.

Example 3.3. *$\lambda = (3, 2, 2)$, then*

$$T = \begin{array}{|c|c|c|} \hline 1 & 4 & 7 \\ \hline 2 & 5 \\ \cline{1-2} 3 & 6 \\ \cline{1-2} \end{array}$$

Take the column stabilizer is

$$C_r = S_{\mu_1} \times S_{\mu_2} \times \cdots,$$

when the μ_i are the column lengths. Note λ has more than n-parts, thus $\mu_i > n$.

$$\Rightarrow S^\lambda \mathbb{C}[S - r] a_T b_T,$$

where b_T is the column anti symmetrizer $b_T = \sum_{\sigma \in C_T} \operatorname{sgn}(\sigma)\sigma$,

$$\mathcal{S}_\lambda(V) = \operatorname{Hom}_{S_r}(S^\lambda, V^{\otimes r}) = \operatorname{Hom}_{S_r}(\mathbb{C}[S_r] a_T b_T, V^{\otimes r}) \cong a_T b_T V^{\otimes r} \overset{\text{Equation (3.1.2)}}{=} a_T(\wedge^{\mu_1} V \otimes \wedge^{\mu_2} V \otimes \cdots).$$

We have,

$$\dim \wedge^{\mu_i} V = \binom{\dim V}{\mu_i} = \binom{n}{\mu_i},$$

and 0i f $\mu_i > n$.

If $\mu_i > n$ then $\wedge^\mu V = 0 \Rightarrow \mathcal{S}_\lambda(V) = 0$. □

Definition 3.1. *Let $X_1, \cdots, X_r \in \operatorname{Mat}_{n\times n}(\mathbb{C})$, and $c = (i_1, \cdots, i_p) \in S_r$. We define,*

$$\phi_c(X_1, \cdots, X_r) = \operatorname{tr}(X_{i_1}, \cdots, X_{i_p}).$$

Lemma 3.3. *Let $X_1, \cdots, X_r \in \operatorname{Mat}_{n\times n}$ and $\sigma \in S_r$ where $\sigma = c_1 \cdots c_k$ decoposes into disjoint cycles. We have,*

$$\operatorname{tr}(\underbrace{\sigma \circ (X_1 \otimes \cdots \otimes X_r)}_{\operatorname{End}_{\mathbb{C}}(V^{\otimes r})}) = \prod_{j=1}^{k} \phi_{c_i}(X_1, \cdots, X_r). \tag{3.1.3}$$

Proof. Note that W is a finite dimensional vector space, thus ,

$$\operatorname{End}_{\mathbb{C}}(W) \cong W \underset{\mathbb{C}}{\otimes} W^*, A \mapsto w \otimes f, \operatorname{tr}(A) \to f(w).$$

For a fixed σ, Equation (3.1.3) is multilinear in $X_1, \cdots, X_r$. It suffices to prove this for $X_i = u_i \otimes f_i \in V \otimes V^* = \operatorname{End}(V)$.

$$\begin{aligned}
\sigma \circ (X_1 \otimes \cdots X_r)(v_1 \otimes \cdots \otimes v_r) &= \sigma(f_1(v_1)u_1 \otimes \cdots \otimes f_r(v_r)u_r), \\
&= \prod_{i=1}^{r} f_i(v_i)\sigma(u_1 \otimes \cdots \otimes u_r), \\
&= \prod_{i=1}^{r} f_i(v_i) u_{\sigma^{-1}(1)} \cdots \otimes u_{\sigma^{-1}(r)}.
\end{aligned}$$

Hence,

$$\sigma \circ (X_1 \otimes \cdots \otimes X_r) = \overbrace{((u_{\sigma^{-1}(1)} \otimes f_1) \otimes \cdots \otimes (u_{\sigma^{-1}(r)} \otimes f_r))}^{\in (V \otimes V^*)^r = \operatorname{End}(V)^{\otimes r} = \operatorname{End}(V^{\otimes r})}.$$

Thus we get,

$$\mathrm{tr}(\sigma\circ(X_1\otimes\cdots\otimes X_r)) = f_1(u_{\sigma^{-1}(1)})\cdots f_r(u_{\sigma^{-1}(r)}).$$

On the other hand, for $c_i = (j_1,\cdots,j_l)$,

$$\phi_{c_i}(X_1,\cdots,X_r) = \mathrm{tr}(X_{j_1},\cdots,X_{j_l}) = \mathrm{tr}((u_{j_1}\otimes f_{j1})\circ\cdots\circ(u_{j_l}\otimes f_{j_l})).$$

We have,

$$\mathrm{End}(W)\times\mathrm{End}(W)\to\mathrm{End}(W), (w\otimes w^*)\times(w\otimes w^*)\to w\otimes w^*, (v\otimes f, u\otimes g)\mapsto f(u)v\otimes g.$$

Applying this, we get,

$$\begin{aligned}\phi_{c_i}(X_1,\cdots,X_r) &= \mathrm{tr}(f_{j_1}(u_{j_2})f_{j_2}(u_{j_3})\cdots f_{j_{l-1}}(u_{j_l})u_{j_1}\otimes f_{j_l}),\\ &= f_{j_1}(u_{j_2})\cdots f_{j_{l-1}}(u_{j_l})f_{j_l}(u_{j_1}),\\ &= \prod_{p=1}^{l} f_{j_p}(u_{c_i(j_p)}),\\ &= \prod_{p=1}^{l} f_{j_p}(u_{\sigma(j_p)}).\end{aligned}$$

Thus we get,

$$\prod\phi_{c_i}(X_1,\cdots,X_r) = \prod_{i=1}^{r} f_i(u_{\sigma(i)}) = \mathrm{tr}(\sigma\circ(X_1\otimes\cdots\otimes X_r)).$$

□

Proposition 3.3. *Denote the character of the representation $S_\lambda(V)$ by*

$$\rho_\lambda\in\mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n} = \mathcal{O}(T)^{\S_n} = \mathbb{C}\otimes\Lambda_n[e_n^{-1}].$$

We have the following statements.

1). $S_\lambda\neq\{0\}\Leftrightarrow\lambda$ has at most n-parts.

2). $\rho_\lambda = s_\lambda$.

Proposition 3.4. *Note that* 2) ⇒ 1)*, since $\rho_\lambda = s_\lambda\neq 0$ thus $S_\lambda(V)\neq\{0\}$.*

Let

$$X = \begin{pmatrix} t_1 & 0 & \cdots & 0\\ 0 & t_2 & \cdots & 0\\ \vdots & \ddots & \ddots & \vdots\\ 0 & 0 & \cdots & t_n.\end{pmatrix}\in T = \left\{\begin{pmatrix} * & 0 & \cdots & 0\\ 0 & * & \cdots & 0\\ \vdots & \ddots & \ddots & \vdots\\ 0 & 0 & \cdots & *.\end{pmatrix}\right\}$$

Then

$$\mathrm{tr}(X) = \sum_{i=1}^{n} t_i, \mathrm{tr}(X^p) = \sum_{i=1}^{n} t_i^p.$$

If $c = (i_1, \cdots, i_p)$ then

$$\phi_c(X, \cdots, X) = \mathrm{tr}(X^p) = \sum_{i=1}^{n} t_i^p = \psi_p(t_1, \cdots, t_n).$$

This means that if σ has cycle type $\mu = (\mu_1 \geq \mu_2 \geq \cdots)$ (ie. $\sigma = c_1 c_2 \cdots$, where c_i is a cycle of length μ_i).

$$\begin{aligned}\mathrm{tr}(\sigma^{-1} \circ (X \otimes \cdots \otimes X)) &\overset{\text{Lemma 3.3}}{=} \phi_{c_1}(X, \cdots, X)\phi_{c_2}(X, \cdots, X) \cdots, \\ &\overset{\text{By definition}}{=} \psi_\mu(t_1, \cdots, t_n), \\ &\overset{\text{By definition}}{=} \sum_{\lambda \text{ with at most } n\text{-parts}} c_\lambda(\mu) s_\lambda(t_1, \cdots, t_n).\end{aligned}$$

On the other hand, we have a decomposition of tensor spaces.

$$V^{\otimes r} = \bigoplus_{\lambda \text{ has at most } n\text{-parts}} S_\lambda(V) \boxtimes S^\lambda,$$

as $\mathrm{GL}_n \times S_r$-representations. This leads us to that

$$\mathrm{tr}(\sigma^{-1} \circ (X \otimes \cdots \otimes X)) = \sum_{\lambda \text{ has at most } n\text{-parts}} \rho_\lambda(t_1, \cdots, t_n) \underbrace{\chi_\lambda(\mu)}_{= c_\lambda(\mu)}.$$

Finally, we get,

$$\forall \mu, 0 = \sum_{\lambda \text{ has at most } n\text{-parts}} c_\lambda(\mu)(\rho_\lambda - s_\lambda).$$

Set,

$$C = (c_\lambda(\mu))_{\lambda \text{ has at most } n\text{-parts, } \mu \text{ is any partition}}.$$

we get,

$$C(\rho_\lambda - s_\lambda)_{\lambda \text{ has at most } n\text{-parts}} = 0.$$

Now we show that C has full rank. Indeed,

$$\begin{aligned}&\tilde{C} = (c_\lambda \times \textit{number of conjugacy classes of } \mu)_{\lambda \text{ at most } n\text{-parts and } \mu \text{ arbitrary}}. \\ &\Rightarrow \tilde{C}^T C = \sum_\mu c_\lambda(\mu) c_\lambda(\mu)(\textit{number of conjugacy classes of } \mu) = r! \,\mathrm{id}\,.\end{aligned}$$

Thus we conclude $\rho_\lambda = s_\lambda$ for all λ with at most n-parts.

Recall that for partitions $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_)$ and $\mu = (\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n)$ with at most n-parts. We have defined,

$$\lambda + \mu = (\lambda_1 + \mu + 1 \geq \cdots \geq \lambda_n + \mu_n).$$

Note that

$$\lambda + \mu \neq \lambda \oplus \mu.$$

Lemma 3.4.

$$\mathcal{S}_\lambda(V) \otimes \det = \mathcal{S}_{\lambda+(1^n)}(V).$$

Proof. Let us take,

$$T_1 \cdots T_2 A_{\lambda+\rho} = \sum_\sigma \operatorname{sgn}(\sigma) T_1^{\lambda-(n-1)+1} T_2^{\lambda_2-(n-2)+1} \cdots T_n^{\lambda_n+1} = A_{\lambda+(1^n)+\rho}.$$

That is,

$$T_1 T_2 \cdots T_n s_\lambda = T_1 \cdots T_n \frac{A_{\lambda+\rho}}{V} = \frac{A_{\lambda+(1^n)+\rho}}{V} = s_{\lambda+(1^n)}.$$

Since

1. det has character $T_1 \cdots T_n$,
2. $\mathcal{S}_\lambda(V)$ has character s_λ,
3. $\mathcal{S}_\lambda(V)$ has character $s_{\lambda+(1^n)}$.

We conclude that the character of $\mathcal{S}_\lambda(V) \otimes \det$ is

$$= T_1 \cdots T_n s_\lambda = s_{\lambda+(1^n)}.$$

□

Theorem 3.2. *THe following is a full collection of (pairwise non-isomorphic)* GL_n*-representations.*

$$\mathcal{S}_\lambda(V) \otimes \det{}^{\otimes m},$$

where λ *is a partition with at most* $n-1$ *parts and* $m \in \mathbb{Z}$*.*

In particular, there is a bijection,

$$\{\textit{sequence of integers } a_1 \geq \cdots \geq a_n \textit{ in } \mathbb{Z}\} \overset{1:1}{\leftrightarrow} \operatorname{Irr}(\mathrm{GL}_n).$$

given by

$$(a_1, \cdots, a_n) \mapsto \mathcal{S}_{(a_1-a_n, a_2-a_n, \cdots, a_{n-1}-a_n)} \otimes \det{}^{\otimes a_n}.$$

Proof. Any irreducible representation is of the form $\mathcal{S}_\lambda(V) \otimes \det^k$ where λ has at most n-parts. Write

$$\lambda = \mu + (\lambda_n^n),$$

such that μ has at most $n-1$-parts. Then we get,

$$\mathcal{S}_\lambda(V) \otimes \det{}^k \overset{\text{Lemma 3.4}}{=} \mathcal{S}_\mu(V) \otimes \underbrace{\det \otimes \cdots \otimes \det}_{\lambda_n\text{-times}} \otimes \det{}^{\otimes k} = \mathcal{S}_\mu(V) \otimes \det{}^{\lambda_n+k}.$$

Remains to show that

$$\mathcal{S}_\lambda(V) \otimes \det \cong \mathcal{S}_\mu(V) \otimes \det{}^{l-m} \Leftrightarrow \lambda = \mu, l = m.$$

Without the loss of generality, we assume $m = 0$ and $l \geq 0$. Then,

$$\mathcal{S}_\lambda(V) \cong \mathcal{S}_m u(V) \otimes \det{}^{\otimes l} \cong \mathcal{S}_{\mu+(l(1^n))}(V).$$

Hence

$$\mathcal{S}_\lambda = \mathcal{S}_{\mu+l(1^n)} \Rightarrow \lambda = \mu + l(1^n).$$

□

Now we classify $\mathrm{Irr}(\mathrm{SL}_n)$.

Lemma 3.5. *Restriction along* $\mathrm{SL}_n \hookrightarrow \mathrm{GL}_n$ *induces an isomorphism,*

$$\mathcal{O}(\mathrm{SL}_n)^{\mathrm{SL}_n} \cong \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}/(\det -1) = \mathbb{C}[T_1, \cdots, T_n]^{S_n}/(e_n - 1).$$

Proof. There is a short exact sequence of vector spaces,

$$0 \longrightarrow (\det -1)\mathcal{O}(\mathrm{GL}_n) \longrightarrow \mathcal{O}(\mathrm{GL}_n) \longrightarrow \mathcal{O}(\mathrm{SL}_n) \longrightarrow 0$$

Using that SL_n is linearly reductive, we conclude $(\cdot)^{\mathrm{SL}_n}$ is exact by the exercise sheet. We thus get,

$$0 \longrightarrow ((\det -1)\mathcal{O}(\mathrm{GL}_n))^{\mathrm{SL}_n} \longrightarrow (\mathcal{O}(\mathrm{GL}_n)))^{\mathrm{SL}_n} \longrightarrow (\mathcal{O}(\mathrm{SL}_n))^{\mathrm{SL}_n} \longrightarrow 0$$

is exatct. Thus any $\lambda I_n \in \mathrm{GL}_n$ acts trivially on $\mathcal{O}(\mathrm{GL}_n)$, and $\mathrm{GL}_n = \langle \mathrm{SL}_n, \lambda I_n \rangle$. Thus we conclude,

$$\mathcal{O}(\mathrm{GL}_n)^{\mathrm{SL}_n} = \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}.$$

If $f \in \mathcal{O}(\mathrm{GL}_n)$ such that $(\det -1)t \in \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}$. Then $f \in \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}$ since $\det \in \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}$. Thus we obtain,

$$((\det -1)\mathcal{O}(\mathrm{GL}_n))^{\mathrm{GL}_n} = (\det -1)\mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}.$$

Thus we get,

$$\mathcal{O}(\mathrm{SL}_n)^{\mathrm{SL}_n} = \mathcal{O}(\mathrm{GL}_n)^{\mathrm{SL}_n}/((\det -1)\mathcal{O}(\mathrm{GL}_n))^{\mathrm{SL}_n} = \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}/(\det -1)\mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}.$$

□

Remark 3.4 (Warning)**.**

$$\mathrm{GL}_n / Z(\mathrm{GL}_n) \ncong \mathrm{SL}_n .$$

Theorem 3.3. *Any irreducible* GL_n*-representation remains irreducible upon restriction to* SL_n*. The following is a complete set of (pairwise non-isomorphic)* SL_n *representations.*

$$S_\lambda(V) \quad \lambda \text{ has at most } n-1\text{-parts}.$$

Proof. Let $\mathcal{M}$ be an irreducible GL_n-representation. Let $\mathcal{N} \subseteq \mathcal{M}$ which is a n SL_n-subrepresentation. For any $\lambda \in \mathbb{C}$, the element $\lambda I_n \in \mathrm{GL}_n$ is central. By Schur's lemma, $\lambda \cdot I_n$ acts by multiplying with some $\tilde{\lambda} \in \mathbb{C}^*$ on $\mathcal{M}$. Therefore, $\mathcal{N}$ is stable under λI_n.

Since $\mathrm{GL}_n = \langle \mathrm{SL}_n, \lambda I_n \mid \lambda \in \mathbb{C}^\times \rangle \Rightarrow \mathcal{N}$ is a GL_n-subrepresentation. Thus we conluder $\mathcal{N} = \{0\}$ or $\mathcal{M}$. Thus $\mathcal{M}$ is irreducible as an SL_n-representation.

In particular,

$$V^{\otimes r} = \bigoplus_{\substack{\lambda \vdash r \\ \text{at most } n\text{-parts}}} S_\lambda(V)^{\otimes \dim S^\lambda} \boxtimes S^\lambda,$$

as both GL_N-representations and SL_n-representations.

Since any irreducible SL_n representation appears in some $V^{\otimes r}$, we conclude any irreducible SL_n-representation is of the form S_λ.

If $\lambda_n > 0$, we can write

$$\lambda = \mu + \lambda_n(1^n),$$

for some μ with at most $n-1$-parts. Thus,

$$S_\lambda(V) = S_\mu(V) \otimes \det^{\otimes \lambda_n} \cong S_\mu(V),$$

as SL_n-reprensetation. But notice that $\det^{\otimes \lambda_n} \cong \mathrm{triv}$ as SL_n-representation.

To see that $s_\lambda(V) \ncong S_\mu(V)$, if $\lambda \neq \mu$ both at most $n-1$ parts, suffices to show that they have distinct characters in

$$\mathcal{O}(\mathrm{SL}_n)^{\mathrm{SL}_n} = \mathbb{C}[T_1, \cdots, T_n]^{S_n}/(e_n - 1).$$

It is an exercise that

$$\{S_\lambda \mid \lambda \text{ at most } n-1\text{-parts}\}$$

is linearly independent in $\mathcal{O}(\mathrm{SL}_n)^{\mathrm{SL}_n}$. □

Let us compare the above theorem with $\mathfrak{sl}_n$. Recall that

$$\mathrm{Irr}^{\mathrm{fd}}(\mathfrak{sl}_n) \overset{1:1}{\Leftrightarrow} \{\text{dominant integral weights}\}.$$

Let $\eta := \{a_1E_{11} + a_2E_{22} + \cdots a_nE_{nn} \mid \sum a_i = 0\}$, which is called the Cartan subalgebra. We also had,

$$\eta^* := \mathrm{Span}_{\mathbb{C}}\{E_{11}^*, \cdots, E_{n-1n-1}^*\}.$$

4 Lie algebras of an algebraic groups

Recall the following definition.

Definition 4.1. *A weight of a Lie algebra $\mathfrak{sl}_n$ is an element $\lambda \in \eta^*$. It is called dominant if*

$$\forall i = 1, \cdots, n-1, \lambda(E_{ii} - E_{i+1i+1}) \in \mathbb{Z}_{\geq 0}.$$

If we have,

$$\lambda = \lambda_1 E_{11}^* + \cdots + \lambda_{n-1}E_{n-1n-1}^*.$$

Then λ is dominant if and only if all the coefficients are positive integers and

$$\lambda_1 \geq \cdots \geq \lambda_{n-1} \geq 0.$$

Or equivalently, $\lambda = (\lambda_1, \cdots, \lambda_{n-1})$ is a partition with at most $n-1$-parts.

$$\mathrm{Irr}(\mathfrak{sl}_n) \overset{1:1}{\leftrightarrow} \{\text{partition of at most } n-1\text{-parts}\} \overset{1:1}{\leftrightarrow} \mathrm{Irr}(\mathrm{SL}_n).$$

The goal is to find a connection between algebraic groups and Lie algebras explaining this phenomena.

Throughout the section, we let k to be an algebraically closed field with any characteristic. Let G be an algebraic group.

Definition 4.2. *A k-derivation on $\mathcal{O}(G)$ is a k-linear map $\delta : \mathcal{O}(G) \to \mathcal{O}(G)$ satisfying Leipniz rule. Explicitly,*

$$\delta(fg) = f\delta(g) + g\delta(f).$$

Remark 4.1. *For any derivation δ, we have $\delta(1) = 0$. If δ_1, δ_2 are k-derivations on $\mathcal{O}(G)$ then their commutator $[\delta_1, \delta_2] = \delta_1 \circ \delta_2 - \delta_2 \circ \delta_1$ is again a derivation.*

Remark 4.2. *Recall, for any $g \in G$, we can define an algebra automorphism*

$$\lambda_g : \mathcal{O}(G) \to \mathcal{O}(G), f(\cdot) \mapsto f(g^{-1}(\cdot)).$$

Definition 4.3. *A derivation $\delta : \mathcal{O}(G) \to \mathcal{O}(G)$ is said to be left-invariant if*

$$\forall g \in G, \lambda_g \circ \delta = \delta \circ \lambda_g.$$

We denote the set of all left-invariant derivations to be $\mathcal{L}(G)$.

Remark 4.3. *For left-invariant derivations δ_1, δ_2, their commutator $[\delta_1, \delta_2]$ is again left invariant. Thus $\mathcal{L}(G)$ equipped with the bracket $[\cdot, \cdot]$ is a Lie algebra.*

Definition 4.4. *For an algebraic group G, $\mathcal{L}(G)$ is the Lie algebra of G.*

We will examine following questions,

1. What is the dimension of $\mathcal{L}(G)$?
2. If $f : G \to G'$ is a homomorphism of algebraic groups, can we get $\mathcal{L}(G) \to \mathcal{L}(G')$?
3. What does $\mathcal{L}(G)$ know about $\mathrm{Rep}(G)$?

Example 4.1. *Let $G = \mathbb{G}_a$. Recall $\mathcal{O}(G) = k[T]$. We would like to have a left-invariant function $\delta : k[T] \to k[T]$. Examine,*

$$\delta(T^n) = T\delta(T^{n-1}) + T^{n-1}\delta(T).$$

By induction, we get,

$$\delta(T^n) = n \cdot \delta(T)T^{n-1}.$$

By left-invariance, we have, for any $x \in G$,

$$\lambda_x \circ \delta(T) = \delta \circ \lambda_x \circ (T) = \delta(T - x) = \delta(T).$$

In another words, $\delta(T)$ is stable under all substitutions $T \mapsto T - X$. Thus we conclude $\delta(T)$ is a constant.

We conclude $\delta(T)$ is a multiple of $\frac{\partial}{\partial T}$ we conclude,

$$\mathcal{L}(G) = k \cdot \frac{\partial}{\partial T}.$$

Definition 4.5. *Let X be a variety. For any $x \in X$, we defined $\mathfrak{m}_x := \{f \in \mathcal{O}(X) \mid f(x) = 0\}$. The Zariski tangent space is defined as*

$$T_x X = \mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k).$$

Example 4.2. *Let $X = \mathbb{A}^n$, and $x = (x_1, \cdots, x_n) \in \mathbb{A}^n$, then*

$$\mathfrak{m}_x = (T_1 - x_1, \cdots, T_n - x_n).$$

Then we observe,

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \text{ is a n-dimensional with basis } T_1 - x_1, \cdots, T_2 - x_2.$$

That is

$$T_x\mathbb{A}^n = (\mathfrak{m}_x, \mathfrak{m}_x^2)^*$$

is also n-dimensional with dual basis,

$$\alpha_1, \cdots \alpha_n, \alpha_i(T_i + x_i + \mathfrak{m}_x^2) = \delta_{ij}.$$

Thus we can write,

$$\alpha_i(f) = \frac{\partial f}{\partial T_i}(x).$$

We denote this basis by,

$$\frac{\partial}{\partial T_1}(x), \cdots, \frac{\partial}{\partial T_n}(x).$$

Definition 4.6. *Let $f : X \to Y$ be a morphism of varieties with corresponding algebra homomorphism, $f^\# : \mathcal{O}(Y) \to \mathcal{O}(X)$. If $x \in X$, then the preimage $(f^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{m}_{f(x)}$. In particular, $f^\#$ induces a linear map,*

$$\mathfrak{m}_{f(x)}/\mathfrak{m}_{f(x)}^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2.$$

Taking duals yeilds a map,

$$df_x : \mathfrak{m}_x/\mathfrak{m}_x^2 = T_xX \to T_{f(x)}Y = \mathfrak{m}_{f(x)}/\mathfrak{m}_{f(x)}$$

Lemma 4.1. *Let $X \subseteq \mathbb{A}^n$ be adefined by $f_1, \cdots, f_m \in \mathcal{O}(\mathbb{A}^n)$. Then the inclusion,*

$$\iota : X \to \mathbb{A}^n,$$

identifies the space T_xX via $d\iota_x : T_xX \to T_x\mathbb{A}^n$ with the linear subspace T_xX of $T_x\mathbb{A}^n$ given by

$$\ker(J(x)),$$

where

$$J(x) = J_{f_1,\cdots,f_m}(x)\left(\frac{\partial f_j}{\partial T_i}(x)\right)_{\substack{1\le i\le n\\ 1\le j\le m}}.$$

Proof. Let $\mathfrak{m}_x \subseteq \mathcal{O}(\mathbb{A}^n)$ and $\overline{\mathfrak{m}_x} = \mathfrak{m}_x/I(X) \subseteq \mathcal{O}(X)$. We have an exact sequence.

$$k^n \xrightarrow{e_i \to f_i + \mathfrak{m}_x^2} \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \overline{\mathfrak{m}_x}/\overline{\mathfrak{m}_x}^2 \longrightarrow 0$$

Taking the duals, we get,

$$0 \longrightarrow T_xX \longrightarrow T_x\mathbb{A}^n \xrightarrow{\alpha^*} k^n$$

Note that $T_x\mathbb{A}^n$ has the basis

$$\left\{ \frac{\partial}{\partial T_1}(x), \cdots, \frac{\partial}{\partial T_n} \right\}.$$

And in k^n we have the dual basis corresponding to the standard basis $\{e_i\}$. We have,

$$\alpha^* \left(\frac{\partial}{\partial T_i}(x) \right)(e_j) = \left(\frac{\partial}{\partial T_i}(x) \right) \underbrace{(\alpha(e_j))}_{f_i + \mathfrak{m}_x^2} = \frac{\partial f_j}{\partial T_i}(x).$$

Thus we obtain,

$$\alpha^* \left(\frac{\partial}{\partial T_i}(x) \right) = \sum_j \frac{\partial f_j}{\partial T_i}(x) e_j^*.$$

Therefore, α^* is the linear map given by $J(x)$. □

Lemma 4.2. *Let $\varphi : \mathbb{A}^n \to \mathbb{A}^m$ be such that*

$$\varphi = (\varphi_1(T_1, \cdots, T_n), \cdots, \varphi_m(T_1, \cdots, T_n)).$$

Then

$$d\varphi_x : T_x\mathbb{A}^n \to T_x\mathbb{A}^m,$$

is the linear given by the Jacobi matrix,

$$J(x) = J_\varphi(x) = \left(\frac{\partial \varphi_j}{\partial T_i} \right)_{\substack{1 \le i \le n \\ 1 \le j \le m}}.$$

Proof. The map $\mathfrak{m}_{\varphi(x)}/\mathfrak{m}_{\varphi(x)}^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2$ is given by

$$T_j - \varphi(x)_j + \mathfrak{m}_{\varphi(x)}^2 \mapsto \varphi_j - \varphi(x)_j + \mathfrak{m}_x^2.$$

Hence,

$$d\varphi_x \left(\frac{\partial}{\partial T_i}(x) \right)(T_j - \varphi(x)_j + \mathfrak{m}_{\varphi(x)}^2) = \left(\frac{\partial}{\partial T_i}(x) \right)(\varphi_j - \varphi(x)_j + \mathfrak{m}_x^2) = \frac{\partial \varphi_j}{\partial T_i}(x).$$

Thus we obtain,

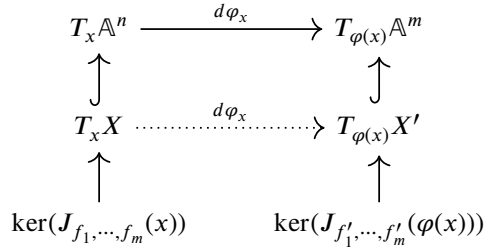
$$d\varphi_x \left(\frac{\partial}{\partial T_i} \right) = \sum_j \frac{\partial \varphi_j}{\partial T_i}(x) \cdot \frac{\partial}{\partial T_j}(\varphi(x)).$$

We conclude that $d\varphi_x$ is the linear map given by $J_\varphi(x)$. □

This can be used to describe any morphisms of affine varieties on tangent spaces.

let $X \subseteq \mathbb{A}^n$ bedefined by $f_1, \cdots, f_m$ and $X' \subseteq \mathbb{A}^m$ bedefined by $f_1', \cdots, f_m'$. Any morphism,

$$\begin{array}{ccc} \mathbb{A}^n & \xrightarrow{\varphi} & \mathbb{A}^m \\ \uparrow & & \uparrow \\ X & \xrightarrow[\varphi]{} & X' \end{array}$$



Theorem 4.1. *Let $f : X \to Y$ be a morphism of irreducible varieties.*

1). $\forall x \in X$, we have $T_xX \geq \dim X$.

2). There si adense open subset $U \subseteq X$ such that for all $x \in U$, $\dim T_xX = \dim X$.

3). Let $\mathrm{char}(k) = 0$, If $f : X \to Y$ is dominant there is a dense open subset $U \subseteq X$ such that $df_x : T_xX \to T_{f(x)}Y$ is surjective for all $x \in U$.

Proof. Omitted. □

Example 4.3. *Let p be a prime number $f : \mathbb{A}^1 \to \mathbb{A}^1$ be such that*

$$f(T) = T^p.$$

At $T = 1$, we have,

$$df_1 = p,$$

(ie the multiplication by p). In particular, if $\mathrm{char}(k) = p$, then df_1 is the 0-map. Similarly, $df_x = 0$ for all $x \in \mathbb{A}^1$ if $\mathrm{char}(k) = p$.

Definition 4.7. *A point $x \in X$ is called simple if*

$$\dim T_xX = \dim X.$$

We call X smooth if all points of X are simple.

Example 4.4. $\mathrm{Spec}(k[x, y]/(xy))$ *is not smooth since the origin is not simple.*

Lemma 4.3. *Let G be an algebraic group. Any $x \in G$ is simple. In particular G is smooth.*

Proof. For any $g \in G$, we get an isomorphism of varieties by

$$G \ni x \mapsto gx \in G.$$

Thus we get,

$$\forall x, g \in G, T_xG \xrightarrow{\sim} T_{gx}G$$

Thus all $x \in G$, we have $\dim T_xG$ is the same. By Theorem 4.1 2). there eixsts $x \in G$ with

$$T_xG = \dim G,$$

thus all points of G are simple. □

Theorem 4.2. *Let G be an algebraic group. There is an isomorphism of vector spaces,*

$$\theta : \mathcal{L}(G)\delta \mapsto (f \mapsto \delta(f)(e)) \in T_eG,$$

with the inverse,

$$\eta : T_eG \ni X \mapsto \delta_X = [g \mapsto X(\lambda_{g^{-1}}(f) - f(g))] \in \mathcal{L}(G).$$

Proof. We first show that θ is well-defined. To do so, need to show that $\theta(\delta) = \delta(\cdot)(e)$ vanishes on $\mathfrak{m}_e^2$. Let $f, g \in \mathfrak{m}_e$, then,

$$\theta(\delta)(fg) = \delta(fg)(e) = (f\delta(g) + \delta(f)g)(e) = 0,$$

since $f, g \in \mathfrak{m}_e$, they vanishe on e. Secondly, we show that η is well-defined. We will do it by deriving,

$$\lambda_{g^{-1}} - f(g) \in \mathfrak{m}_e.$$

We already have,

$$(\lambda_{g^{-1}}(f) - f(g))(e) = f(g) - f(g) = 0.$$

Now we check that δ_x is a derivation,

$$\begin{aligned}
\eta_x(f_1f_2)(g) &\overset{\text{def}}{=} x(\lambda_{g^{-1}}(f_1f_2) - f_1(g)f_2(g)),\\
&= x(\lambda_{g^{-1}}(f_1)\lambda_{g^{-1}}(f_2) - f_1(g)f_2(g)),\\
&= x(\underbrace{(\lambda_{g^{-1}}(f_1) - f_1(g))}_{\in\mathfrak{m}_e}\underbrace{(\lambda_{g^{-1}}(f_2) - f_2(g))}_{\in\mathfrak{m}_e} + f_1(g)(\lambda_{g^{-1}}(f) - f_1(g)) + (\lambda_{g^{-1}}(f) - f_1(g))f_2(g)).\\
\Rightarrow &(\lambda_{g^{-1}}(f_1) - f_1(g))(\lambda_{g^{-1}}(f_2) - f_2(g)),\\
\eta_x(f_1f_2)(g) &= x(f_1(g)(\lambda_{g^{-1}}(f_2) - f_2(g)) + (\lambda_{g^{-1}}(f) - f_1(g))f_2(g)),\\
&= f_1(g)\cdot x(\lambda_{g^{-1}}(f_2) - f_2(g) + (\lambda_{g^{-1}}(f_1) - f_1(g))f_2(g)),\\
&= f_1(g)\delta_x(f_2)(g) = \delta_x(f_1)(g)f_2(g),\\
\Rightarrow &\delta(f_1f_2) = f_1\delta_x(f_2) + \delta(f_1)f_2.
\end{aligned}$$

We have δ is left invariant. Indeed,

$$\begin{aligned}
(\lambda_{g_2}\circ\delta_x)(f)(g_1) &= \delta_x(f)(g_2^{-1}g_1),\\
&= x(\lambda_{g_1^{-1}g_2}(f) - f(g_2^{-1}g_1)),\\
&= x(\lambda_{g_1^{-1}}(\lambda_{g_2}(f)) - \lambda_{g_2}(f)(g_1)),\\
&= \delta_x(\lambda_{g_2}(f))(g_1).\\
\Rightarrow &\lambda_{g_2}\circ f = f\circ\lambda_{g_2}.
\end{aligned}$$

We now show that θ and η are inverse to each other.

$$\begin{aligned}
\eta(\theta(\delta))(f)(g) &= \theta(\delta)(\lambda_{g^{-1}}(f) - f(g)),\\
&= \delta(\lambda_{g^{-1}}(f) - f(g))(e),\\
&\overset{\delta(1)=0}{=} \delta(\lambda_{g^{-1}}(f))(e),\\
&\overset{\delta \text{ is left invariant}}{=} \lambda_{g^{-1}}(\delta(f))(e),\\
&= \delta(f)(g).\\
\Rightarrow &\eta(\theta(\delta)) = \delta.
\end{aligned}$$

Let $x \in (\mathfrak{m}_e/\mathfrak{m}_e^2)^*$ and $f \in \mathfrak{m}_e$.

$$\begin{aligned} \theta(\eta(x))(f) &= \eta(x)(f)(e), \\ &= x(\\ lambda_{g^{-1}}(f) - f(g)), & \\ &= x(f). \\ \Rightarrow \theta(\eta(x)) &= x. \end{aligned}$$

This finishes the proof. □

From now on, we denote the Lie algebra of G by $\mathfrak{g}$ where $\mathfrak{g}$ inherits the Lie bracket from $\mathcal{L}(G)$.

Corollary 4.1. *We have*

$$\dim_k \mathfrak{g} = \dim G.$$

In particular, $\mathfrak{g}$ *is finite-dimensional.*

Proof. Since G is smooth we have,

$$\dim_k T_g G = \dim G,$$

for any $g \in G$. □

Recall that GL_n has a ring of functions,

$$\mathcal{O}(\mathrm{GL}_n) = k[T_{ij}]_{1 \le i,j \le n}[y]/(\det y - 1).$$

Lemma 4.4. $\mathfrak{gl}_n$ *is* n^2*-dimensional with basis* $\frac{\partial}{\partial T_{ij}}(I_n) \in (\mathfrak{m}_e/\mathfrak{m}_e^2)^*$ *where, for* $f \in \mathbb{C}[T_{ij}], m \in \mathbb{Z}_{0\ge}$,

$$\frac{\partial(f \det^{-m})}{\partial T_{ij}}(I_n) := \frac{\partial f}{\partial T_{ij}}(I_n) - m f(I_n)\delta_{ij}.$$

Proof. We need to first show that $\frac{\partial}{\partial T_{ij}}(I_n)$ is well-defined.

Note that

$$\frac{\partial \det}{\partial T_{ij}}(I_n) = \delta_{ij},$$

since we have,

$$\det = T_{11} \cdots T_{nn} + \sum_{\sigma \in \mathfrak{S}_n, \sigma \neq 1} \mathrm{sgn}(\sigma) T_{1\sigma(1)} \cdots T_{n\sigma(n)}.$$

Then the right hand side contains at least two factors of T_{ij} if $i \neq j$.

$$\begin{aligned}
\frac{\partial(f \det^a) \det^{-a-m}}{\partial T_{ij}}(I_n) &= \frac{\partial(f \det^a)}{\partial T_{ij}}(I_n) - (a+m) f(I_n)\delta_{ij}, \\
&= \left(\frac{\partial f}{\partial T_{ij}}(I_n) \overset{a}{\det} + f \frac{\partial \det^a}{\partial T_{ij}} \right)(I_n) - (a+m) f(I_n)\delta_{ij}, \\
&= \left(\frac{\partial f}{\partial T_{ij}}(I_n) \overset{a}{\det} + f a \overset{a-1}{\det} \frac{\partial \det}{\partial T_{ij}} \right)(I_n) - (a+m) f(I_n)\delta_{ij}, \\
&= \frac{\partial f}{\partial T_{ij}}(I_n) + a f(I_n) f_{ij} - (a+m) f(I_n)\delta_{ij}, \\
&= \frac{\partial f}{\partial T_{ij}}(I_n) - m f(I_n)\delta_{ij}, \\
&= \frac{\partial f \det^m}{\partial T_{ij}}(I_n).
\end{aligned}$$

We know that

$$\dim \mathfrak{gl}_n = \dim \mathrm{GL}_n = n^2.$$

Let $x := \sum \lambda_{ij} \frac{\partial}{\partial T_{ij}}(I_n) = 0$. Then

$$\underbrace{x(T_{ab})}_{=0} = \sum_{\lambda_{ij}} \frac{\partial T_{ab}}{\partial T_i j}(I_n) = \lambda_{ab}.$$

In particular,

$$\Rightarrow \lambda_{ab} = 0 \forall a, b,$$

$\Rightarrow$The $\frac{\partial}{\partial T_{ij}}(I_n)$ are linearly independent. By the dimension argument, we have $\frac{\partial}{\partial T_{ij}}(I_n)$ are the basis.

Remains to show that each $\frac{\partial}{\partial T_{ij}}(I_n)$ vanishes on $\mathfrak{m}_{I_n}^2$. Let $f \det^{-a}, g \det^{-b} \mathfrak{m}_{I_n}$. Then,

$$\begin{aligned}
\frac{\partial(f \det^{-a})(g \det^{-b})}{\partial T_{ij}}(I_n) &= \frac{\partial(fg) \det^{-a-b}}{\partial T_{ij}}(I_n), \\
&= \frac{(fg)}{\partial T_{ij}}(I_n) - (a+b) f(I_n) g(I_n)\delta_{ij}, \\
&= \left(f \frac{\partial g}{\partial T_{ij}} + \frac{\partial f}{\partial T_{ij}} \right)(I_n), \\
&= 0.
\end{aligned}$$

□

Lemma 4.5. *The Lie bracket* $[\cdot,\cdot] : \mathfrak{gl}_n \times \mathfrak{gl}_n \to \mathfrak{gl}_n$*, which we identify it with* $\mathfrak{gl}_n \cong \mathrm{Mat}_{n\times n}$ *and* $\frac{\partial}{\partial T_{ij}} \mapsto E_{ij}$ *is the commutator bracket on* $\mathrm{Mat}_{n\times n}$*.*

Proof. Recall the isomorphism $\eta : T_e \operatorname{GL}_n \to \mathcal{L}(\operatorname{GL}_n)$. So we compute the bracket on $\eta\left(\frac{\partial}{\partial T_{ij}}(I_n)\right)$. We have,

$$\begin{aligned}
\eta\left(\frac{\partial}{\partial T_{ij}}(I_n)\right)(T_{ab})(A) &= \frac{\partial}{\partial T_{ij}}(I_n)(\lambda_{A^{-1}}(T_{ab}) - T_{ab}(A)),\\
&= \frac{\partial(\lambda_{A^{-1}}(T_{ab}) - A_{ab})}{\partial T_{ij}},\\
&= \frac{\partial(\sum_l A_{al}T_{lb} - A_{ab})}{\partial T_{ij}},\\
&= A_{ai}\delta_{jb},\\
\Rightarrow \eta\left(\frac{\partial}{\partial T_{ij}}(I_n)\right)(T_{ab}) &= T_{ai}\delta_{jb}.
\end{aligned}$$

On $\mathcal{L}(\operatorname{GL}_n)$ the Lie bracket is the commutator thus,

$$\begin{aligned}
\eta\left(\frac{\partial}{\partial T_{ij}}(I_n)\right) \circ \eta\left(\frac{\partial}{\partial T_{lm}}(I_n)\right)(T_{ab}) &= \eta\left(\frac{\partial}{\partial T_{ij}}(I_n)\right)(T_{al}\delta_{b,m}),\\
&= T_{ai}\delta_{lj}\delta_{bm},\\
&= \delta_{jl}\eta(\left(\frac{\partial}{\partial T_{im}}(I_n)\right))(T_{ab}),\\
\Rightarrow \left[\eta\left(\frac{\partial}{\partial T_{ij}}(T_n)\right), \eta\left(\frac{\partial}{\partial T_{lm}}(T_n)\right)\right](T_{ab})&\\
= \left(\delta_{j,l}\eta\left(\frac{\partial}{\partial T_{im}}(I_n)\right) - \delta_{i,m}\eta\left(\frac{\partial}{\partial T_{im}}(I_n)\right)\right)(T_{ab}).&
\end{aligned}$$

Since any derivation ofn $\mathcal{O}(\operatorname{GL}_n)$ is uniquely determined by the values on the T_{ab}(From Leibniz rule, $0 = \delta(\det \det^{-1}) = \det \delta(\det^{-1}) + \delta(\det)\det^{-1}$), we have,

$$\Rightarrow \left[\eta\left(\frac{\partial}{\partial T_{ij}}(I_n)\right), \eta\left(\frac{\partial}{\partial T_{lm}}(I_n)\right)\right] = \delta_{jl}\frac{\partial}{\partial T_{im}}(I_n) - \delta_{im}\eta\left(\frac{\partial}{\partial T_{lj}}\right).$$

This is precisely the commutator bracket on $\operatorname{Mat}_{n\times n}$ where,

$$[E_{ij}, E_{lm}] = \delta_{j,l}E_{i,m} - \delta_{i,m}E_{l,j}.$$

□

For any morphism of algebraic groups, $\varphi : G \to G'$, we write,

$$d\varphi = d_e\varphi : \underbrace{\mathfrak{g}}_{=T_eG} \to \underbrace{\mathfrak{g}'}_{=T_eG}.$$

Proposition 4.1. *Let $\varphi : G \to G'$ be a morphism of algebraic grups then $d\varphi : \mathfrak{g} \to \mathfrak{g}'$ is a morphism of Lie algebras.*

Proof. Let $tildedφ : \mathcal{L}(G) \to \mathcal{L}(G')$ be the map induced by $d\varphi$ via η, θ. In other words,

$$\begin{array}{ccc} T_eG = \mathfrak{g} & \xrightarrow{d\varphi} & T_eG' = \mathfrak{g}' \\ \theta\uparrow & & \downarrow\eta \\ \mathcal{L}(G) & \xrightarrow[\tilde{d\varphi}]{} & \mathcal{L}(G') \end{array}$$

We want to show that $\tilde{d\varphi}$ is a morphism of Lie algebras. Consider,

$$\begin{aligned} \tilde{d\varphi}(\delta)(f)(g) &= \eta(d\varphi(\theta(\delta)))(f)(g), \\ &\overset{\text{def of } \eta}{=} d\varphi(\theta(\delta))(\lambda_{g^{-1}}(f) - f(g)), \\ &\overset{\text{def}}{=} \theta(\delta)\circ\varphi^{\#}(\lambda_{g^{-1}}(f) - f(g)), \\ &\overset{\text{def}}{=} \theta(\delta)(\lambda_{g^{-1}}(f)\circ\varphi - f(g)), \\ &= \delta(\lambda_{g^{-1}}(f)\circ\varphi - f(g))(e), \\ &= \delta(\lambda_{g^{-1}}(f)\circ\varphi)(e). \end{aligned}$$

Claim 4.1.1.

$$\tilde{d\varphi}(\delta)(f)\circ\varphi = \delta(f\circ\varphi).$$

Proof of the claim.

$$\begin{aligned} \tilde{d\varphi}(\delta)(f)\circ\varphi(g) &= \delta(\lambda_{\varphi(g)^{-1}}(f)\circ\varphi)(e), \\ &= \delta(\lambda_{g^{-1}}(f\circ\varphi))(e), \\ &\overset{\delta \text{ is left inv.}}{=} \lambda_{g^{-1}}(\delta(f\circ\varphi))(e), \\ &= \delta(f\circ\varphi)(g). \end{aligned}$$

□

Hence,

$$\begin{aligned} [\delta, \delta'](f\circ\varphi) &= \delta(\delta'(f\circ\varphi)) - \delta'(\delta(f\circ\varphi)), \\ &\overset{\text{Claim}}{=} \delta(\tilde{d\varphi}(\delta')(f)\circ\varphi) - \delta'(\tilde{d\varphi}(\delta)(f)\circ\varphi), \\ &= \tilde{d\varphi}(\delta)\circ\tilde{d\varphi}(\delta')(f)\circ\varphi - \tilde{d\varphi}(\delta')\circ\tilde{d\varphi}(\delta)(f)\circ\varphi, \\ &= [\tilde{d\varphi}(\delta), \tilde{d\varphi}(\delta)](f)\circ\varphi. \end{aligned}$$

Thus we get,

$$\begin{aligned} \tilde{d\varphi}([\delta, \delta'])(f)(g) &= [\delta, \delta'](\lambda_{g^{-1}}(f)\circ\varphi)(e), \\ &= [\tilde{d\varphi}(\delta), \tilde{d\varphi}(\delta')](\lambda_{g^{-1}}(f))\circ\varphi(e), \\ &= [\tilde{d\varphi}(\delta), \tilde{d\varphi}(\delta')](\lambda_{g^{-1}}(f))(e), \\ &\overset{\text{left inv.}}{=} \lambda_{g^{-1}}[\tilde{d\varphi}(\delta), \tilde{d\varphi}(\delta')](f)(g), \\ \Rightarrow \tilde{d\varphi}([\delta, \delta']) &= [\tilde{d\varphi}(\delta).\tilde{d\varphi}(\delta')]. \end{aligned}$$

□

Remark 4.4. *If $G \subseteq \mathrm{GL}_n \subseteq \mathrm{Mat}_{n\times n}$ is the vanishing locus of $f_1, \cdots, f_m \in \mathcal{O}(\mathrm{Mat}_{n\times n})$, then $\mathfrak{g} = \ker(\underbrace{J_{f_1,\cdots,f_m}(I_n)}_{\subseteq T_{I_n}\mathrm{Mat}_{n\times n}})$. And by Proposition 4.1, the Lie bracket on $\mathfrak{g} \subseteq \mathfrak{gl}_n$ is again the commutator bracket.*

Example 4.5. *Let $G = \mathrm{SL}_n = V(\det -1)$, recall that ,*

$$\frac{\partial \det}{\partial_{T_{ij}}} = \delta_{ij}.$$

Thus we have,

$$\mathfrak{sl}_n = \{A \in \mathfrak{gl}_n \mid \sum \delta_{ij} A_{ij} = 0\} = \{A \in \mathfrak{gl}_n \mid \mathrm{tr}(A) = 0\}.$$

Example 4.6. *Let $J \in \mathrm{GL}_n$, and we set $G(J) = \{A \in \mathrm{Mat}_{n\times n} \mid A^T J A = J\}$. Then we have,*

1). J is symmetric, then $\mathbb{G}(J) = O_n$,

2). J is skew-symmetric, then $\mathbb{G}(J) = \mathrm{Sp}_{2n}$.

$G(J)$ is cut out by the following equation. At the (i,l)-entry.

$$\sum_{j,k=1}^{n} A_{ji} J_{jk} A_{kl} = J_{il} \quad (1 \le i, l \le n).$$

Now,

$$\frac{\partial(\sum_{j,k} T_{ji} J_{jk} T_{kl} - J_{il})}{\partial T_{ab}}(I_n) = \delta_{bi} J_{al} + \delta_{bl} J_{ia}.$$

Thus the Lie algebra of $\mathfrak{g}(J) \subseteq \mathfrak{gl}_n = \mathrm{Mat}_{n\times n}$ is a cut out by the equations

$$\sum_{a,b} A_{a,b}(\delta_{ib} J_{al} + J_{ia}\delta_{lb}) = \sum_{a} (A_{ia} J_{al} + A_{al} J_{ia}) = (A^T J)_{il} + (J A)_{il}$$

Or another words, this is equivalent to $A^T J + J A = 0$. THus we get,

$$\mathfrak{g}(J) = \{A \in \mathrm{Mat}_{n\times n} \mid A^T J + J A = 0\}.$$

In particular, the orthogonal group O_n has Lie algebra $\mathfrak{so}_n$ and Sp_n has Lie algebra $\mathfrak{sp}_{2n}$

Theorem 4.3. *Let $\imath : G \to G, g \mapsto g^{-1}$ be the inversion .Then we have,*

$$d\imath : \mathfrak{g} \to \mathfrak{g}, d\imath(x) = -x.$$

Let $\mu : G \times G \to G, (g,h) \mapsto gh$ be the multiplication, then we have,

$$d\mu : \mathfrak{g} \oplus \mathfrak{g} \to \mathfrak{g}, (x,y) = x + y.$$

Proof. We first prove the second assertion. Since the differential of

$$G \overset{j=(\mathrm{id},e)}{\to} G \times G,$$

is given by

$$d_j(x) = (d\,\mathrm{id}(x), de(x))$$

Since any map factoring through $\{e\} \hookrightarrow G$ must have the tangent space 0 since,

$$G \longrightarrow \{e\} \longrightarrow G$$

then

$$T_eG \longrightarrow \{0\} \longrightarrow T_eG, \qquad T_eG \xrightarrow{0} T_eG$$

Thus the map d_j is given by,

$$d_j(x) = (x, 0).$$

Note that $\mu \circ j(g) = \mu(g, e) = g$ thus we have $\mu \circ j = \mathrm{id}$. THus for any $x \in \mathfrak{gl}_n$, we have,

$$x = d_\mu(d_j(x)) = d\mu(x, 0).$$

By the additivity, we obtain,

$$d_\mu((x, y)) = d_\mu((x, 0) + (0, y)) = d_\mu(x, 0) + d_\mu(0, y) = x + y.$$

Now, we prove the first assertion. The following composition is the constant map e,

$$G \xrightarrow{\Delta = g \mapsto (g,g)} G \times G \xrightarrow{(\mathrm{id}, \iota)} G \times G \xrightarrow{\mu} G$$

The following composition is the zero map,

$$\mathfrak{g} \xrightarrow{\Delta} \mathfrak{g} \oplus \mathfrak{g} \xrightarrow{(\mathrm{id}, d_\iota)} \mathfrak{g} \oplus \mathfrak{g} \xrightarrow{d\mu} \mathfrak{g}$$

Hence, we have, for any $x \in \mathfrak{g}$,

$$0 = d\mu \circ (\mathrm{id}, d_\iota) \circ \Delta(x) = d\mu(x, d_\iota(x)) = x + d_\iota(x).$$

Thus we conclude $d_\iota(x) = -x$. □

We assume from now on in this section that $\mathrm{Rep}(G), \mathrm{Rep}(\mathfrak{g})$ are all finite dimensional representations of G and $\mathfrak{g}$, respectively.

Definition 4.8. *Let G be an algebraic group and V be a G-representation. Let $\rho : G \to \mathrm{GL}(V)$ be the corresponding morphism of algebraic groups. Taking differentials, we get a morphism of Lie algebras.*

$$d\rho : \mathfrak{g} \to \mathfrak{gl}(V).$$

We denote this corresponding representation of $\mathfrak{g}$ by $d(V) \in \mathrm{Rep}(\mathfrak{g})$, which is a finite dimensional representation.

Example 4.7. *Let $V = \mathrm{triv}$, be the trivial representation, in other words,*

$$\rho : G \to \mathrm{GL}_1, g \mapsto e.$$

Thus from the discussion in Theorem 4.3, we obtain $d\rho = 0$. In other words, $d(\mathrm{triv}_G) = \mathrm{triv}_{\mathfrak{g}}$.

Next goal is to extend this to a functor $d : \mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g})$.

Lemma 4.6. *The assingment $\mathrm{Rep}(G) \ni V \mapsto d(V) \in \mathrm{Rep}(\mathfrak{g})$ is compatible with tensor products and duals. That is to say,*

$$d(V \otimes W) \cong d(V) \otimes d(W), d(V^*) \cong d(V)^*.$$

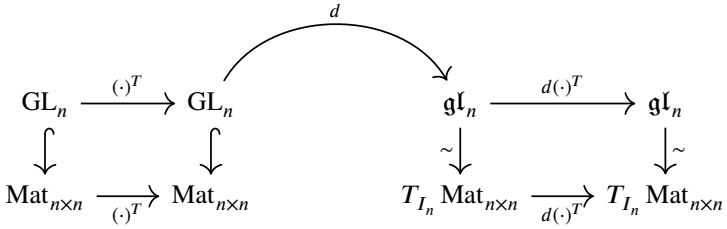
Proof. Let $\rho : G \to \mathrm{GL}_n$ be a representation on G. The dual representation is the composition of ,

$$\rho^* : G \xrightarrow{\iota} G \xrightarrow{\rho} \mathrm{GL}_n \xrightarrow{(\cdot)^T} \mathrm{GL}_n .$$

On tangent spaces, this becomes,

$$d\rho^* : \mathfrak{g} \xrightarrow{d\iota} \mathfrak{g} \xrightarrow{d\rho} \mathfrak{gl}_n \xrightarrow{d(\cdot)^T} \mathfrak{gl}_n.$$

Note that $d(\cdot)^T = (\cdot)^T$, since,



$(\cdot)^T : \mathrm{Mat}_{n\times n} \to \mathrm{Mat}_{n\times n}$ is linear since $d(\cdot)^T = (\cdot)^T$ on $\mathfrak{gl}_n$ is

$$d\varphi^*(x) = -xd\rho(x)^T.$$

This is the dual representation of $d\rho : \mathfrak{g} \to \mathfrak{gl}_n$.

The tensor product of $\rho : G \to \mathrm{GL}(V), \rho' : G \to \mathrm{GL}(W)$ is the following map,

$$G \xrightarrow{\Delta} G \times G \xrightarrow{(\rho,\rho')} \mathrm{GL}(V) \times \mathrm{GL}(W) \xrightarrow{\alpha:(f,g)\mapsto f\otimes g} \mathrm{GL}(V \otimes W)$$

Taking differentials we get,

$$\mathfrak{g} \xrightarrow{\Delta} \mathfrak{g} \times \mathfrak{g} \xrightarrow{(d\rho,d\rho')} \mathfrak{gl}(V) \oplus \mathfrak{gl}(W) \xrightarrow{d\alpha} \mathfrak{gl}(V \otimes W)$$

We have a commutative diagram,

$$\begin{array}{ccc} \mathrm{GL}(V) \times \mathrm{GL}(W) & \xrightarrow{\qquad\qquad\alpha\qquad\qquad} & \mathrm{GL}(V \otimes W) \\ \downarrow & & \downarrow \\ \mathrm{End}(V) \times \mathrm{End}(W) & \xrightarrow[\tilde{\alpha}:(f,g)\mapsto f\otimes g]{} & \mathrm{End}(V) \otimes \mathrm{End}(W) = \mathrm{End}(V \otimes W) = \mathrm{End}(V \otimes W) \end{array}$$

Let $j : \mathrm{End}(V) \to \mathrm{End}(V) \times \mathrm{End}(W), f \mapsto (f, \mathrm{id}_W)$. Then $\tilde{\alpha} \circ j$ is a linear map

$$(f \mapsto f \otimes \mathrm{id}_W).$$

Since we have,

$$d(\tilde{\alpha} \circ j)(f) = f \otimes \mathrm{id}_W = d\tilde{\alpha}(d_j(f)) = d\tilde{\alpha}(f, 0).$$

(Since $j = (\mathrm{id}_{\mathrm{End}(V)}, c_{id_W})$ where $c : \mathrm{End}(V) \to \mathrm{End}(W), f \mapsto \mathrm{id}_W$ whose differential is of course 0). Similarly

$$d\tilde{\alpha}(0, g) = \mathrm{id}_V \otimes g.$$

Thus we obtain,

$$d\tilde{\alpha}(f, g) = d\tilde{\alpha}((f, 0) + (0, g)) = f \otimes \mathrm{id}_W + \mathrm{id}_V \otimes f.$$

Taking the differential in the commutative diagram above, we get,

$$\begin{array}{ccc} \mathfrak{gl}(V) \oplus \mathfrak{gl}(W) & \xrightarrow{d\alpha} & \mathfrak{gl}(V \otimes W) \\ \sim\downarrow & & \downarrow\sim \\ T_e \mathrm{End}(V) \times T_e \mathrm{End}(W) & \xrightarrow[d\tilde{\alpha}]{} & T_e \mathrm{End}(V \otimes W) \end{array}$$

Let $\rho \otimes \rho'$ be the tensor product representation the, we get,

$$d(\rho \otimes \rho')(x) = d\alpha(\rho(x), \rho'(x)) = \rho(x) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \rho'(x).$$

By definition, we get,

$$d(\rho \otimes \rho') = d(\rho) \otimes d(\rho').$$

□

Proposition 4.2. *The assignment,*

$$d : \mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g}), V \mapsto d(V), f \mapsto d(f) = f,$$

is a functor.

Proof. Need to chevk if $f \in \mathrm{Hom}_G(V, W)$ then $f \in \mathrm{Hom}_{\mathfrak{g}}(d(V), d(W))$.

Note that $f \in \mathrm{Hom}_{\mathbb{C}}(V, W)$ is G-linear if and only if f lies in $f \in (W \otimes V^*)^G = \mathrm{Hom}_G(V, W)$. Since

$$d(\mathrm{triv}_G) = \mathrm{triv}_{\mathfrak{g}},$$

we have,

$$d(\mathcal{M}^G) \subseteq d(\mathcal{M})^{\mathfrak{g}},$$

for all G-representations $\mathcal{M}$. If $f \in \mathrm{Hom}_{\mathbb{C}}(V, W)$ is G-equivariant, then

$$\begin{aligned} f \in (V^* \otimes W)^G \Rightarrow f &\in d((V^* \otimes W)^G), \\ &\subseteq d(V^* \otimes W)^{\mathfrak{g}}, \\ &= (d(V)^* \otimes d(W))^{\mathfrak{g}}, \\ &= \mathrm{Hom}_{\mathbb{C}}(d(V), d(W))^{\mathfrak{g}}, \\ &= \mathrm{Hom}_{\mathfrak{g}}(d(V), d(W)). \end{aligned}$$

□

Remark 4.5. *Thus $d : \mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g})$ is always a faithful functor. However, it is not always full.*

Example 4.8. *Let $G = \mathbb{G}_m$ and consider the one-dmensional $\mathbb{G}_m$-representation k_p where $t \in \mathbb{G}_m$ acts by multiplication with t^p. Then $d\rho = p(\cdot) : k = \mathfrak{gl}_1 \to k = \mathfrak{gl}_1$ is multiplication by p. Thus if $\mathrm{char}(k) = p$, then $d(k_p) \cong \mathrm{triv}$. Therefore, we have,*

$$\dim \mathrm{Hom}_{\mathfrak{g}}(d(\mathrm{triv}), d(k_p)) = 1.$$

But triv, k_p are non-isomorphic irreducible representations of $\mathbb{G}_m$. That is

$$\mathrm{Hom}_{\mathbb{G}_m}(\mathrm{triv}, k_p) = 0.$$

Lemma 4.7. *Let V be a G-representation and $v \in V$. Then the differential of the map,*

$$\varphi : G \to V, g \mapsto gv,$$

is

$$d\varphi_e : \mathfrak{g} \to V, x \mapsto xv.$$

Proof. Note that φ factors through

$$\begin{array}{ccccc} G & \xrightarrow{\rho} & \mathrm{GL}(V) & \xrightarrow{(\cdot)v} & V \\ & & \downarrow & & \downarrow \sim \\ & & \mathrm{End}(V) & \xrightarrow[(\cdot)v \text{ which is linear}]{} & V \end{array}$$

The differentials

$$\mathfrak{g} \xrightarrow{d\rho} \mathfrak{gl}(V) \xrightarrow{(\cdot)v} V, x \mapsto xv.$$

□

Proposition 4.3. *Let $k = \mathbb{C}$. If G is connected and V is a G-representation, then $V^G = d(V)^{\mathfrak{g}}$.*

Proof. We already know that $V^G \subseteq d(V)^{\mathfrak{g}}$. Let $v \in d(V)^{\mathfrak{g}}$ and consider,

$$\varphi : G \to \overline{Gv}, g \mapsto g_v.$$

This is a dominant morphism. From Theorem 4.1, there is a $g \in G$ such that

$$d\varphi_g : T_g G \to T_{gv}\overline{Gv},$$

is surjective. By G-equivariance. We have,

$$d\varphi_e : \mathfrak{g} \to T_v\overline{Gv},$$

is also surjective. Clearly $T_v\overline{Gv} \subseteq \mathfrak{g}$. By Lemma 4.7, we have,

$$d\varphi_e(x) = xv = 0.$$

That is $d\varphi_e = 0$ is surjective, thus $T_v\overline{Gv} = 0$.

By G-equivariance, $T_x\overline{Gv} = 0$ for all $x \in Gv$. Since Gv is dense in $\overline{Gv}$ and by Theorem 4.1, there is a dense open subset $U \subseteq \overline{Gv}$ such that for all $x \in U$,

$$\dim T_x\overline{Gv} = \dim \overline{Gv}.$$

By density, $Gv \cap U \neq \emptyset$. Let $x \in Gv \cap U$ them

$$\dim \overline{Gv} = \dim T_x\overline{Gv} = \dim\{0\} = 0.$$

Thus we conclude Gv is a finite set of points. But G is connected thus G is irreducible. Since the image of an irreducible set Gv is also irreducible thus $\overline{Gv}$ is also irreducible. We conclude $\overline{Gv} = \{v\}$. Thus we conclude $v \in V^G$. □

Theorem 4.4. *Let $k = \mathbb{C}$ and G is a connected algebraic group. Then the functor,*

$$d : \mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g}),$$

is fully-faithful. That is this functor induces an isomorphism,

$$\mathrm{Hom}_G(V, W) \xrightarrow{d} \mathrm{Hom}_{\mathfrak{g}}(d(V), d(W)),$$

for any $V, W \in \mathrm{Rep}(G)$.

Proof. We have,

$$\mathrm{Hom}_G(V, W) = (V^* \otimes W)^G \overset{\text{Prop 4.3}}{=} d(V^* \otimes W) = (d(V)^* \otimes d(W))^{\mathfrak{g}} = \mathrm{Hom}_{\mathfrak{g}}(d(V), d(W)).$$

□

Remark 4.6. *Theorem 4.4 never woks if G is not connected. For example, consider a finite group G. Then $\mathfrak{g} = \{0\}$, and*

$$d : \mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g}) = (\mathrm{Vect})$$

is just the forgetful functor. Thus d is not full (there are linear maps which are not G-linear).

Definition 4.9. *A connected algebraic group G over $k = \mathbb{C}$, is called semisimple if $\mathfrak{g}$ is semisimple.*

Example 4.9. *We've already seen that $\mathrm{SL}_n, O_n, \mathrm{Sp}_{2n}$ have semisimple Lie algebras, $\mathfrak{sl}_n, \mathfrak{so}_n, \mathfrak{sp}_{2n}$. Thus $\mathrm{SL}_n^\circ, O_n^\circ, \mathrm{Sp}_{2n}^\circ$ are all semisimple. Unfortunately O_n is not connected. But we've shown in the exercise sheet that SO_n is connected. Thus $\mathrm{SL}_n, \mathrm{SO}_n, \mathrm{Sp}_{2n}$ are all semisimple.*

Theorem 4.5. *Let G be a connected semisimple algebraic group over $k = \mathbb{C}$. Then $\mathrm{Rep}(G)$ is semisimple and $V \in \mathrm{Rep}(G)$ is irreducible if and only if $d(V)$ in $\mathrm{Rep}(\mathfrak{g})$ is irreducible.*

Proof. For semisimplicity, we need to show any short exact sequence,

$$0 \longrightarrow V_1 \longrightarrow V_2 \longrightarrow V_3 \longrightarrow 0$$

in $\mathrm{Rep}(G)$ splits.

Since $\mathrm{Rep}(\mathfrak{g})$ is semisimple, the sequence,

$$0 \longrightarrow d(V_1) \longrightarrow d(V_2) \xrightarrow{f} d(V_3) \longrightarrow 0$$

splits , so there is $g \in \mathrm{Hom}_{\mathfrak{g}}(d(V_2), d(V_3))$ such that $g \circ f = \mathrm{id}_{V_2}$. But we have,

$$\mathrm{Hom}_{\mathfrak{g}}(d(V_2), d(V_3)) = \mathrm{Hom}_{g}(V_2, V_3).$$

Thus $g \in \mathrm{Hom}_G(V_2, V_3)$, thus the short exact sequence in the beginning splits.

For the latter part, we see that, if $V \in \mathrm{Rep}(G)$ is irreducible then $\dim \mathrm{End}(G)(V) = 1$ if and only if $\dim \mathrm{End}_{\mathfrak{g}}(d(V)) = 1$ if and only if $d(V)$ is irreducible. □

Let T be a torus and V a T-representation. We have,

$$\mathcal{O}(T) \ni \chi_{T,V}(t) = \mathrm{tr}(t|_V).$$

And we have defined that,

$$X^*(T) = \mathrm{Hom}_{\text{alg-grp}}(T, \mathbb{G}_m).$$

Lemma 4.8. *The inclusion,*

$$X^*(T) \hookrightarrow \mathcal{O}(T),$$

induces an isomorphism,

$$k[X^*(T)] \xrightarrow{\sim} \mathcal{O}(T).$$

Where $k[X^*(T)]$ is a group algebra of $X^*(T)$.

Proof. Both sides transform products of groups into tensor products of algebras. Thus we may assume that $T = \mathbb{G}_m$. We have an isomorphism $Z \ni 1 \mapsto (t \mapsto t) \in X^*(\mathbb{G}_m)$.

$$k[X^*(\mathbb{G}_m)] \xrightarrow{\sim} \mathcal{O}(\mathbb{G}_m) = k[x, x^{-1}],$$

where $X^*(\mathbb{G}_m) \cong \mathbb{Z}$, and the map is given by $\mathbb{Z} \ni 1 \mapsto x \in k[x, x^{-1}]$. □

Hence if,

$$V \cong \bigoplus_{\lambda \in X^*(T)} k_x^* n_x,$$

where k_λ is the one-dimensional T-representatio nwhere t acts by $\chi(t)(\cdot)$. Then

$$\chi_{V,T} = \sum_{\lambda \in X^*(T)} n_\lambda \chi \in k[X^*(T)] \cong \mathcal{O}(T).$$

There is a similar notion for Lie algebras. Let $\mathfrak{t}$ be an abelian Lie algebra and V be a semisimple $\mathfrak{t}$-representation. Then,

$$V = \bigoplus_{\lambda \in \mathfrak{t}^*} k_\lambda^{\oplus n_\lambda},$$

where k_λ is the one-dimensional $\mathfrak{t}$-representation on which $x \in \mathfrak{t}$ acts by $\lambda(t)$.

The character of V is

$$\chi_{\mathfrak{t},V} = \sum n_\lambda e^\lambda \in k[\mathfrak{t}^*],$$

where we denote $e^\lambda e^\mu = e^{\lambda+\mu}$.

Any morphism of algebraic groups,

$$\chi : T \to \mathbb{G}_m$$

gives rise to a linear map,

$$\mathfrak{t}^* \ni d\lambda : \mathfrak{t} \to k.$$

The functor can be written as,

$$\mathrm{Irr}(\mathrm{Rep}(G)) \ni k_\chi \overset{d}{\mapsto} d(k_\chi) = k_{d(\chi)} \in \mathrm{Irr}(\mathrm{Rep}(\mathfrak{t})).$$

Thus,

$$V = \bigoplus_{\lambda \in X^*(T)} k_\lambda^{n_\lambda} \Rightarrow d(V) = \bigoplus_{\lambda \in X^*(\mathfrak{t})} k_{d(\lambda)}^{n_\lambda}.$$

There is a map,

$$d : k[X^*(T)] \ni \chi \mapsto d(\chi) \in k[\mathfrak{t}^*],$$

and

$$d(\chi_{T,V}) = \sum_{\lambda \in X^*(T)} n_\lambda(\lambda) = \chi_{\mathfrak{t}, d(V)}.$$

Proposition 4.4. *Let λ be a partition with at most n-parts. Then,*

$$d : \mathrm{Rep}(\mathrm{SL}_n) \to \mathrm{Rep}(\mathfrak{sl}_n),$$

sends $S^\lambda(V)$ to the irreducible highest weight representation of weight

$$\lambda = \sum \lambda_i E_{ii}^* \in \mathfrak{t}^*,$$

where $\mathfrak{t}$ is the standard Cartan subalgebra of $\mathfrak{sl}_n$, (ie. all diagonal entries in $\mathfrak{sl}_n$).

Proof. Let $T \subseteq \mathrm{SL}_n$ be the diagonal torus which has Lie algebra $\mathfrak{t}$. Note that,

$$T_i \in \mathbb{C}[T_1, \cdots, T_n]/(e_n - 1) \cong \mathcal{O}(T).$$

Picks at the i-th entry in T. Then, $d(T_i)$ picks at the i-th coordinate in $\mathfrak{t}$, thus $d(T_i) = E_{ii}^*$. We have,

$$\chi_{T, S^\lambda(V)} \overset{\text{Prop 3.3}}{=} s_\lambda = \frac{A_{\lambda+\rho}}{V}.$$

Expanding the right most hand side, we have,

$$= \frac{\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) T_{\sigma(1)}^{\lambda_1 + n - 1} \cdots T_{\sigma(n)}^{\lambda_n}}{\prod_{i<j}(T_i - T_j)}.$$

Apply d to this, we get,

$$\begin{aligned}
\chi_{\mathfrak{t}, d(S^\lambda(V))} &= \frac{\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) e^{(\lambda_1 + n - 1) E^*_{\sigma(1)\sigma(1)} + \cdots + \lambda_n E^{\sigma(n)\sigma(n)}}}{\prod_{i<j}(e^{E_{ii}^*} - e^{E_{jj}^*})}, \\
&= \frac{\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) e^{\sigma(\lambda+\rho)}}{e^\rho \prod_{i<j}(1 - e^{E_{jj}^* - E_{ii}^*})}, \\
&= \frac{\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) e^{\sigma(\lambda+\rho)}}{e^\rho \prod_{\alpha \in \Phi^+}(1 - e^{-\alpha})}.
\end{aligned}$$

Where,

$$\lambda = \lambda_1 E_{11}^* + \cdots + \lambda_n E_{nn}^*,$$
$$\rho = (n-1)E_{11}^* + \cdots + E_{nn}^*,$$
$$\Phi^+ = \{\alpha_{ij} = E_{ii}^* - E_{jj}^* \mid i < j\}.$$

This is the character of the irreducible highest weight representation $L(\lambda)$. Any $\mathfrak{g}$-representation is determined by its $\mathfrak{t}$-character. Thus we have,

$$S^\lambda \cong L(\lambda).$$

□

Let us now consider $G = \mathrm{SO}_{2n}$ over $k = \mathbb{C}$ which is of type D_n. That is

$$\mathrm{SO}_{2n} = \{A \in \mathrm{GL}_n \mid A^* J A = J\},$$

where

$$J = \begin{pmatrix} O & I_n \\ I_n & O \end{pmatrix}.$$

Then

$$\mathfrak{so}_{2n} = \{A \in \mathrm{Mat}_{n\times n} \mid A^* J + J A = 0\}.$$

Let

$$\mathfrak{t} = \left\{ \begin{pmatrix} X & O \\ O & -X \end{pmatrix} \middle| X \in \mathrm{Mat}_{n\times n}, \text{diagonal} \right\},$$

be the standard Cartan subalgebra. Let $e_i = E_{ii}^* = E_{n+i,n+i}^* \in \mathfrak{t}^* (1 \le i \le n)$. We see, $\{e_i \mid 1 \le i \le n\}$ is a basis of $\mathfrak{t}^*$ and the inner product on $\mathfrak{t}^*$ is coming from the killing form is

$$(e_i, e_j) = \delta_{ij}.$$

Then

$$\Phi(\mathfrak{so}_{2n}) = \{\pm e_i \pm e_j \mid i \pm j\}.$$

This defines $(\alpha, \alpha) = 2 \forall \alpha \in \Phi$. Furthermore,

$$\mathbb{Z}\Phi = \{\sum a_i e_i \mid \sum a_i \equiv 0 \mod 2\}.$$

The weight (missing) is

$$\begin{aligned} \bigwedge(\mathfrak{so}_{2n}) &= \{\lambda \in \mathfrak{t}^* \mid \langle \lambda, \alpha \rangle \in \mathbb{Z}, \forall \alpha \in \Phi\}, \\ &= \{\lambda \in \mathfrak{t}^* \mid (\lambda, \alpha) \in \mathbb{Z}, \forall \alpha \in \Phi\}, \\ &= \{\lambda = \frac{1}{2} \sum a_i e_i \in \mathfrak{t}^* \mid a_i \equiv e_j \mod 2 \forall i \ne j, a_i \in \mathbb{Z}\}. \end{aligned}$$

Consider the following insde SO_{2n},

$$T :== \left\{ \begin{pmatrix} D & O \\ O & D^{-1} \end{pmatrix} \middle| D \in \mathrm{GL}_n \text{ is diagonal} \right\}.$$

Then T has Lie algebra $\mathfrak{t}$, $X^*(T)$ has basis $\{\chi_i \mid 1 \le i \le n\}$ where,

$$\chi_i \left(\begin{pmatrix} D & O \\ O & D^{-1} \end{pmatrix} \right),$$

and $d\chi_i = E_{ii}^* = e_i$.

Proposition 4.5. *$d : \mathrm{Rep}(\mathrm{SO}_{2n}) \to \mathrm{Rep}(\mathfrak{so}_{2n})$, is not an equivalence for any $n > 1$.*

Proof. Let $V \in \mathrm{Rep}(G)$, then

$$\chi_{\mathfrak{t},d(V)} = d(\chi_{T,V}) \in k[d(X^*(T))] \subseteq k[\mathfrak{t}^*].$$

However, $d(\chi_i) = e_i$. Therefore,

$$d(X^*(T)) = \bigoplus_{i=1}^{n} \mathbb{Z}e_i.$$

Let $\lambda = \frac{1}{2}(e_1 + \cdots e_n) \in \Lambda(\mathfrak{so}_{2n})$. Since $\lambda \notin d(X^*(T)) \Rightarrow d(V) \ncong L(\lambda)$. For any $V \in \mathrm{Rep}(\mathrm{SO}_{2n}) \Rightarrow d : \mathrm{Rep}(\mathrm{SO}_{2n}) \to \mathrm{Rep}(\mathfrak{so}_{2n})$ is not an equivalence. □

One can prove the following,

Theorem 4.6. *Let G be a connected semisimple algebraic group, $k = \mathbb{C}$. Pick a Cartan subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$, we have,*

1). There is a torus $T \subseteq G$ with Lie algebra $\mathfrak{t}$.

2). The d restricts to an equivalence $d : \mathrm{Rep}(G) \to \{V \,|\, in\, \mathrm{Rep}(\mathfrak{g}) | \chi_{\mathfrak{t},V} \in k[d(X^*(T))]\}$.

Remark 4.7. *It is easyd to check that*

$$d : \mathrm{Rep}(\mathrm{SL}_n) \to \mathrm{Rep}(\mathfrak{sl}_n), d : \mathrm{Rep}(\mathrm{Sp}_{2n}) \to \mathrm{Rep}(\mathfrak{sp}_{2n}),$$

are equivalences but

$$d : \mathrm{Rep}(\mathrm{SO}_n) \to \mathrm{Rep}(\mathfrak{so}_n),$$

is not an equivalence for $n > 3$.

Remark 4.8. *There is a connected semisimple group Spin_n with Lie algebra $\mathfrak{so}_n$ such that*

$$d : \mathrm{Rep}(\mathrm{Spin}_n) \to \mathrm{Rep}(\mathfrak{so}_n),$$

is an equivalence. Moreover, there is a surjective morphism of algebraic groups $\mathrm{Spin}_n \xrightarrow{\alpha} \mathrm{SO}_n$ such that

$$\ker(\alpha) \subseteq Z(\mathrm{Spin}_n), \text{ and } |\ker(\alpha)| = 2.$$